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 TOPICAL REVIEW

Optical Data Center Networking: A Comprehensive Review on Traffic, Switching, Bandwidth Allocation, and Challenges

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ABSTRACT The accelerated growth of data traffic in data centers (DCs) globally is driven by the dominance of multiple emerging data-intensive applications hosted by edge/cloud DCs, such as Internet of Things (IoT), augmented and virtual reality etc. The advent of machine learning/artificial intelligence (ML/AI) and its exploitation in several applications for prediction and optimization purposes has further burdened the traffic conditions within existing DC networks (DCNs) where traffic burstiness and high variability dominates. Most of the existing DCNs worldwide use conventional electrical switches which cannot effectively serve the highly differentiated traffic, while optical switching is currently used by few big-data companies' DCs and only in the core and aggregation DCN tiers. As a result, limited DCN performance in terms of end-to-end delay and bandwidth utilization is provided. Therefore, the investigation on all aspects of optical networking (switching, transmission techniques etc.) in DCs attracts high attention in both academia and industry, aiming to provide high DCN bandwidth utilization, low end-to-end latency (in the order of few μ s), and low data loss. In this paper, a comprehensive review on traffic characteristics, switching techniques, bandwidth allocation schemes and power consumption reduction technologies for next-generation optical DCNs is extensively given. Moreover, trends of optical technology enabled in optical DCNs are comprehensively discussed, putting in the agenda the technical challenges and limitations towards efficient optical DCNs capable of serving the stringent traffic classes demands for guaranteed low completion time of diverse edge/cloud data-intensive applications.

INDEX TERMS Bandwidth allocation, data center, data center network, optical switching, power consumption, traffic classification.

I. INTRODUCTION

The continuous rise of multiple emerging machine learning (ML) and artificial intelligence (AI) applications has caused a tremendous traffic growth in existing data centers (DCs) around the globe. Large datasets are exchanged among servers within DCs to train diverse deep neural networks (DNNs), aiming to provide more accurate ML/AI models. Different parallelism techniques are followed to train the DNNs: either partitioning them across many servers

(model parallelism), or distributing the datasets to multiple servers (data parallelism), or combining the above techniques (hybrid parallelism) [1]. Therefore, an explosive traffic increase overloads the DC networks (DCNs). In this framework, the number of relative computations required in the largest ML/AI training runs met an exponential increase of 3.4-month doubling time during the last decade [2], while this trend is becoming stronger with the advent of distributed data-consuming applications like ChatGPT. In addition, new neuromorphic-based algorithms gradually gain ground in neural network (NN) training models to support several emerging applications, further burdening the data-intensive

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traffic conditions in DCNs [3], [4]. Paradigms of such applications are Google Voice Search, Apple Siri, Amazon Alexa, facial/voice recognition etc.

On the same time, many other popular cloud applications are hosted by existing DCs, often co-existing with ML/AI applications and using the same DCs' resources. In specific, various communication platforms, web-searching machines, social networks, along with diverse on-line gaming, video streaming, Internet-of-Things (IoT), autonomous vehicles, Industry 4.0 [5], and augmented reality/virtual reality/immersive reality (AR/VR/IR) applications [6] provide a tremendously increasing traffic workload that burdens even more the traffic congestion in DCNs. In this context, the initial prediction of Cisco about the rising of Zettabyte era for the DCs traffic globally [7] has already been proven to underestimate the dramatic traffic explosion in DCs during the last years. As Cisco noted with latest predictions, by the end of 2023 IoT and connected-home applications would have represented almost 50% of the machine-to-machine (M2M) connections, while autonomous vehicles compound annual growth rate would have reached more than 30% [8], all served in cloud/edge DCNs. As current trends prove, the edge computing paradigm manages to bring high-performance processing and storage capabilities closer to the end-users and IoT devices than ever before [8], effectively combining the beyond 5G (B5G) and 6G with the optical networking technologies [9]. Finally, the variety of applications and services running in existing DCs provides a high traffic heterogeneity within DCNs in terms of differentiated traffic characteristics like packet/flow size, priority and service class [10], [11].

In a DCN, each end-user application request is divided into a number of separate tasks that are assigned to multiple servers to exploit the distributed processing, memory and storage resources. As both Facebook and Google have already noted for their DCNs, many of the requested computations get over a single server resources capabilities, mainly due to the datasets large size [12], [13]. For this reason, high DCN server-to-server bandwidth -which is referred as bisection bandwidth- is requested by DCs operators to effectively interconnect the servers and to provide guaranteed low end-to-end latency to the multiple data-intensive applications [14]. Despite the technology evolution in hardware accelerators and the use of specific hardware like graphics processing units (GPUs) for intense computation purposes, the response to a DC application request still faces significant delay. For example, today's training tasks for large ML/AI models may last for days, even for weeks [15], [16], [17], while 50% of natural language processing (NLP) models needs more than a day to train, according to measurements at Facebook's DCs [18]. It is noticeable that the large training response time is primarily due to the limited DCN bisection bandwidth and not due to hardware capabilities limitations, since the use of hundreds or even thousands of GPUs is not by itself a sufficient precondition to guarantee the required speed-up [1].

Apart from the traffic growth, another significant parameter that DCs' operators take under consideration for the next-generation DCNs deployment is their energy efficiency, since an exponential increase of global DCs energy footprint is noticed with their expansion [19]. Especially, the DCs energy consumption globally is almost 1% of the total world-wide energy consumption reaching 200 TWh [19], while it is expected to rise to 3-4% by the end of the decade due to ML/AI dominance. Note that only in the USA, the annual DCs energy consumption is more than 90 billion KWh [20]. Hence, in order to reduce the energy footprint, large-scale DC companies gradually decrease the number of active components used in DCNs, while renewable energy procurement is incorporated. The paradigm of Google is representative of this trend, since it has been working to move to optical circuit switches based on micro-electro-mechanical systems (MEMS) for its Jupiter DC by 2022, to provide energy saving up to 41% and transparency increase [21].

A. MOTIVATION

In this context, efficient DCN performance arises as a dominant concern for DCs operators, while its limits investigation attracts the interest of networks research community. DCs operators are mainly interested in providing high-bandwidth DCN whilst guaranteeing low and deterministic end-to-end latency under limited power consumption. Specifically, the relative network engineering challenges aim to respond to the following DCN requirements: 1) high bisection bandwidth for servers interconnection, 2) guaranteed low end-to-end latency considered as network-layer time response in the order of μ s, 3) fast switching time in the order of ns, 4) fast clock data recovery (CDR), 5) effective access control strategy to avoid data loss, 6) DCN reconfigurability under varying traffic conditions, 7) high fault tolerance, 8) low energy consumption, and 9) cost efficiency. According to the above requirements, the network switching technology that can support such switching time constraints is the optical switching technology, as compared to the traditional electrical switching one, since it avoids the power-consuming optical-to-electrical (O/E) and electrical-to-optical (E/O) conversions, removes the electronic data processing delay and provides data transparency, irrespectively of the data rate and format [22], [23]. Towards this direction, large-scale DCs operators -like Google- have already moved to the optical switching era in order to achieve power and performance efficiency [21].

B. RELATED SURVEYS

In this framework, the research interest about large-scale DCs effectiveness to support the constantly increasing number of cloud/edge applications is uninterrupted. In literature, some review studies about the challenges related with different aspects of DCs operation have been introduced. Table 1 presents a comprehensive overview of DCN surveys and their categories according to the (main and complementary) innovative objectives, such as energy consumption, DCN

TABLE 1. Comprehensive overview of DCNs surveys in terms of several objectives.

Main Innovative Objective of Survey	Ref.	Year	Network Technology	Switching Technology	Complementary Objectives of Survey			
					DCN Topology	Traffic Characteristics	Energy Efficiency Perspective	Other
Energy Consumption	[24]	2014	Wired	N/A	✓		Cooling, Computing, Geographical Factors	
	[25]	2014	Wired	Electrical, Optical, Hybrid	✓	✓	DCN Architecture, Virtualization, Cooling, Renewable energy	
	[26]	2021	Wired	N/A			Energy Consumption Modeling	Reliability
DCN Architecture and Topology	[27]	2013	Wired	Electrical, Optical	✓			
	[28]	2014	Wired	N/A	✓			
	[29]	2016	Wired	N/A	✓			
	[30]	2017	Wired	Electrical, Optical, Hybrid	✓		Regional DCs, Power Management	Infrastructures & Operations
	[10]	2018	Wired	Hybrid	✓	✓		
	[31]	2023	Wired	Hybrid	✓			Network Protocols
Re-configurability	[32]	2019	Wired	Optical				
	[33]	2024	Wired	Optical				
DCN Monitoring	[34]	2024	Wired	N/A	✓	✓		Measurement-based Control
Transport in DCNs	[35]	2024	Wired	N/A				Congestion Control
SDN/ OpenFlow	[36]	2022	Wired	N/A				
Virtualization	[37]	2015	Wired	N/A		✓		
Load Balance	[38]	2018	Wired	N/A	✓	✓		
Disaggregation	[39]	2023	Wired	N/A				
Wireless DCNs 60 GHz/FSO	[40]	2016	Wireless		✓			
	[41]	2021	Wireless		✓			
Interconnects/ Switches	[42]	2012	Wired	Hybrid, Optical				
	[48]	2018	Wired	All-Optical				
	[49]	2023	Wired	All-Optical	✓	✓		Components
Bandwidth Allocation/ Optical Switching	This paper		Wired	Optical, Hybrid	✓	✓	Cooling, DCN Architecture, ML/AI Traffic, E/O Switching	SDM for DCNs, Disaggregated DCs
							Integration, Optical Switching Fabric, Optical Transceivers Technology, Optical Routing, Control & Management	

architecture and topology, DCN interconnects and switching technology, wireless DCNs and other objectives. Moreover, in Table 1 the contribution of our study is comparatively given. As it is presented, our study overviews DCNs in terms of traffic characteristics, optical DCNs bandwidth allocation schemes and switching technologies, space division multiplexing (SDM) techniques suitable for optical DCNs, disaggregated DCN architectures and power consumption. In specific according to Table 1, the main objectives of DCN surveys classification is:

1) ENERGY CONSUMPTION

One of the main concerns about DCs effectiveness is perennially their energy footprint. Some of the first surveys about

energy efficiency of traditional electrical switching DCs as well as of the first hybrid (electrical/optical) switching ones are given in [24], [25]. A more recent review of DCs energy consumption and reliability modeling is presented in [26], focusing on reliability indices, models, methodologies, and services.

2) DCN ARCHITECTURES AND TOPOLOGIES

Moreover, one of the first surveys concerning DCN architectures and topologies, was presented a decade ago in [27], discussing several electrical and hybrid DCN topologies and routing protocols. Some years later, other surveys and comparative evaluations of diverse DCN topologies irrespectively of the network switching technology (electrical/optical) were

presented [10], [28], [29], [30], each with different objective view towards a DCN, focusing on network protocols, infrastructures, operations, network maintenance etc. A latest review approach of generic DCN topology design depending on the network component playing a centric operational role (switch/server) is presented in [31].

3) OTHER OBJECTIVES

In addition, diverse surveys have also been published related to other objectives of DCs operation, like DCN reconfigurability [32], [33], DCN monitoring [34], transport in DCNs [35], software-defined networking (SDN) in DCNs along with OpenFlow management capabilities [36], DCN virtualization [37], DCN load balancing [38], and DC disaggregation [39].

4) WIRELESS DCNs

Regarding the technology evolutions in DCNs, although the state-of-the-art (SotA) research focuses on wired optical switching DCNs, the 60 GHz wireless and free-space optics (FSO) technology has also been investigated for the servers interconnection to solve inherit problems such as hot node congestion and cabling complexity, while relative surveys are given in [40] and [41].

5) DCNs INTERCONNECTS AND SWITCHES

To the best of our knowledge, only a few review studies explicitly focus on all-optical DCNs technology. Specifically, one of the first surveys of optical interconnects in DCs was published more than a decade ago in [42] focusing on the hybrid and all-optical switching prototypes for DCNs of that time period projects, such as [43], [44], [45], [46], and [47]. Since then, optical switching technology has significantly evolved. Hence, a more recent review of the advances in optical interconnects/switches and technology for DCN components has been addressed in [48], lightening on optical components like lasers, modulators, multiplexers/demultiplexers (MUXs/DEMUXs) and photodetectors that can be used in DCNs to enhance effectiveness. Finally, in [49] a latest overview of optical switching techniques and challenges for DCNs is provided, focusing on topologies and design requirements.

C. ORIGINAL REVIEW CONTRIBUTION AND STRUCTURE

In the light of the aforementioned, this paper provides an extensive review on the main characteristics of existing DCNs as well as on the dominant technical challenges that will enable next-generation large-scale DCNs to meet the requirements of emerging data-explosive time-sensitive cloud/edge DC applications. Especially as Table 1 shows, we delve into modern DCN traffic characteristics, switching technologies, bandwidth allocation/access control strategies, energy footprint, optical multiplexing technologies like SDM and disaggregated DCN architectures, aiming to derive design directions and trends suitable for performance and power

efficient future optical DCNs. To the best of our knowledge, no other review or survey on optical DCNs combinedly presents all the above optical DCN topics, while none of them reviews bandwidth allocation/access control schemes suitable for optical DCNs.

This paper is organized as follows: Section II gives the main background and basic concepts in existing DCNs while it addresses the trends for near future optical DCN technology evolution. Especially, technology evolutions in DCN optical links are introduced, whilst DCN traffic characteristics are extensively discussed. Moreover, the main design objectives of future efficient all-optical DCNs are comprehensively studied. Section III deals with the switching technologies in DCs, focusing on the optical switching one since it can provide low end-to-end latency and high DCN performance. In specific, different optical technologies used in commercially available and research prototype optical switches for DCNs are presented. Moreover, those optical switching technologies used in current big-data companies' DCNs are lighted along with those promising technologies that are expected to be utilized in next-generation efficient optically switched DCNs. Section IV gives a comprehensive overview on bandwidth allocation schemes and access control algorithms suitable for all-optical DCNs in both intra- and inter-rack optical domain. Different bandwidth allocation techniques are classified according to the performance enhancement provided in terms of end-to-end delay and bandwidth utilization, data contention resolution mechanism, transmission scheme, traffic pattern, medium access control (MAC) protocol etc. Section V focuses on the DCs power consumption since it critically affects their effectiveness. Particularly, different DCN architectures and switching technologies are compared with respect to their energy footprint, providing useful comments for power efficient future optical DCNs design. The main technical challenges and open research topics regarding all-optical DCNs are discussed in Section VI, providing research directions for next-generation efficient optical DCNs capable of effectively serving the heterogeneous traffic of multiple data-intensive DC applications. Finally, Section VII concludes our review.

II. BACKGROUND AND BASIC CONCEPTS

With the tremendous traffic increase in DCNs, there is a constant need for the DCs operators to enhance the DCN performance in order to guarantee the required low task completion time. In this direction, considerable efforts have been put by academia and industry. Based on the latest optical technology evolutions, novel solutions have been proposed for the DCN links bandwidth upgrade, the switching speed-up, and the effectiveness of packet transmission/access control strategies, taking into account the DCN traffic diversity, burstiness and priority classes. Therefore, apart from the energy footprint and cost, other performance parameters such as DCN throughput and end-to-end latency are lately taken under consideration for the DCN performance evaluation.

A. LINKS

Large-scale data companies like Google, Facebook and Microsoft have exploited the wavelength division multiplexing (WDM) technique in DCN links to enhance the total data rate provided [50], [51], [52]. Especially, the coarse WDM (CWDM) technique has been used at the first inter-rack/inter-cluster DCN links based on its low cost technology. CWDM supports in a single fiber the operation of up to 18 wavelengths with a 20 nm distance between them, while they can efficiently work up to few kilometers depending on the exact number of working wavelengths. The main disadvantage of CDWM is the amplification inability because of the large distance between wavelengths, resulting in limited data rate per wavelength. In order to achieve higher data rates using more wavelengths per link and based on improved laser frequency stability technology, the dense WDM (DWDM) technique has been proposed for DCN links [53], since it supports amplification by optical amplifiers. Especially, DWDM can support in a single fiber the operation of many decades of different wavelengths with a smaller distance between them (0.8 and 0.4 nm distance for 48 and 96 wavelengths respectively) without any interference, providing higher data rates per wavelength and longer distance interconnection [54].

The actual data rate achieved per wavelength in a DCN is driven by the pluggable transceiver capabilities as well as by the provided bandwidth of optical interconnects. In a timeline, in 2007 Google first transited to 10 Gb/s optical links in its DCNs for transmissions up to 200 m, using vertical-cavity surface-emitting lasers (VCSELs) with multi-mode fiber (MMF)-based small form-factor pluggable (SFP) transceivers [51]. Almost a decade later, in 2016 Google's Jupiter DCN introduced the use of 40 Gb/s data center interconnects (DCIs) [55]. As the 100 Gb/s optical links were commercially available since 2014, their use in DCNs was adopted almost in 2018 [48]. Since the traffic provided by data-intensive AI applications grow exponentially in DCNs, the requirement for higher link data rates becomes critical. Thus, the 200 Gb/s and 400 Gb/s transceivers standards were presented in 2017 by the IEEE P802.3bs Task Force [56]. In 2019 experimental results on 400 Gb/s DCIs performance were also introduced [57]. Currently the 400 Gb/s technology for DCN links and switching fabric is dominant in big-data DCNs, such as those of Cisco and Google Jupiter [58], [59]. It is evident that the evolution on optical interconnects technology is moving towards the throughput level of Tb/s in order to keep up with the requirements of massively increasing traffic in DCNs. Commercially available switching chips currently support $64 \times 400\text{G}$ ports providing 25 Tb/s switching bandwidth [60], while the prospects for the near future are the support of $32 \times 800\text{G}$ ports. Moreover, since the traffic explosion due to big-data market has propelled DCNs into uncharted territory, requiring extremely high bandwidth capabilities, in 2022 IEEE 802.3ck, P802.3cn, and P802.3df Task Forces introduced relative standards to support the 800G Ethernet technology [61], after the corresponding proposal of Optical Internetworking Forum (OIF). The introduction of

800G transmission capability in DCN links over distances up to 120 km is a milestone and allows for extremely high bandwidth DCNs based on coherent digital signal processor (DSP) technology without the need of costly DWDM systems. The optical market currently provides 800G ZR pluggable modules like [62], while the introduction of 800G switches for large-scale DCNs is gradually being put in the agenda. Therefore, the 800G DC migration becomes a near future reality. According to the market, we expect to achieve 120 km, 500 km and 1200 km cloud interconnection range enabling coherent DSP 800G ZR, 800G ZR+, and 600G ZR pluggable modules respectively [63]. Moreover, based on latest forecasts, 800G ports switches are expected to overpass 400G in demand by 2025, while big-data DC companies such as Google, Meta and Microsoft, as pioneer ones, are forecasted to immediately incorporate the 800G switching technology in their DCNs [64]. Despite the relatively immature stage of current 800G technology for DCNs, market demands have already forced for the 1.6T and 3.2T transmission technology investigation to meet the ever-expanding needs for massive data transmission within DCNs with unparalleled efficiency and reliability. Indeed, in late 2022 the 1.6T standard was presented by IEEE P802.3dj Task Force [65], while the relative for 3.2T is anticipated in the near future. Fig. 1 shows the timeline of DCN link and interconnection data rate evolution. Finally, as the DCNs bandwidth constantly increases according to Fig. 1 and the distance between an end-user/IoT device and the processing network in edge DCs is becoming shorter, data can be transferred faster providing more instantaneous application service.

B. TRAFFIC

Traffic flows and their dependencies are among the main networking challenges when managing cloud and distributed workloads, according to the latest Cisco report [66]. With the massive increase of traffic volume in cloud/edge DCs due to multiple data-intensive applications, efficient DCN design requires a more exploratory investigation of traffic flows characteristics.

The term flow refers to an amount of data that has to be transferred by a source to a destination server within a DCN, related with a specific application task. A data flow is divided into a number of packets by the source server network interface card (NIC), each of which is transmitted independently as a different part of the flow. The adoption of a data flow implies that the respective packets have a sequential and logical correlation among them. Within a DCN, multiple flows originated by different pairs of communicating source-destination servers and corresponding to different application tasks are transferred, requiring diverse levels of quality of service (QoS). Moreover, different flows can be concurrently transmitted by a single source server to different destination servers, while each flow corresponds to a different application task of the source server, adding dynamicity to DCN traffic. As the diversity of cloud/edge applications served by existing

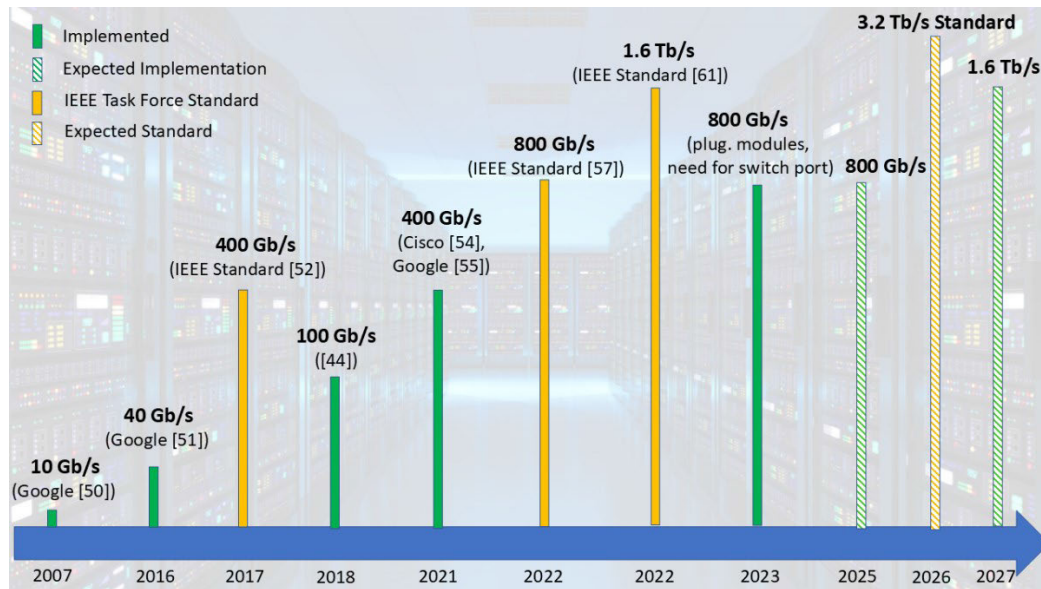


FIGURE 1. Data rate time evolution of DCN links and interconnections.

DCs continuously increases, the flow traffic within DCNs is getting more heterogeneous [67], [68], [69].

Data flows are closely related to the applications running within a DCN. The main applications in modern DCs are listed in [70] and are categorized in the following classes, as shown in Fig. 2: online and real-time applications, computing applications, and storage back-up applications. In the online and real-time applications class belong time-sensitive applications such as video streaming, real-time communication, online gaming, AR/VR/IR, IoT, Industry 4.0 etc. that have strict latency constraints. Hence, the respective traffic flows must be served with the highest priority within the DCN. In the computing applications class belong big-data applications like Hadoop, ChatGPT, ML/AI training etc. that require high availability of computing resources such as central processing units (CPUs)/GPUs and memory. The respective traffic flows are required to face low packet loss at the computing resources; therefore they can be served with medium priority. Finally, in the storage back-up applications class belong applications such as cloud storage, database update, and virtual machine (VM) migration that require high network bandwidth, while they are not sensitive to delays. Hence, the relative traffic flows can be served with low priority.

Within a DCN, traffic patterns strongly depend on the specific applications run in the DC. Knowledge of the DCN traffic patterns is pivotal for the DCN performance optimization in order to design efficient MAC and switching protocols that can ensure the required QoS level in terms of low latency. Nevertheless, for most of the big-data companies, traffic patterns are often assumed as privacy-sensitive and proprietary resulting to insufficient knowledge of traffic characteristics within DCNs. Hence, there are limited real traffic traces available as openly accessible. Therefore, there is a limited

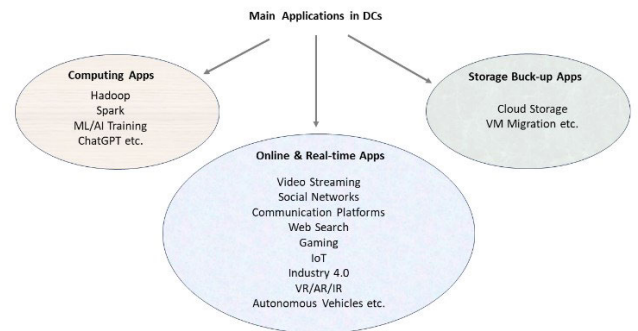


FIGURE 2. Main applications in modern DCs.

body of work related to DCN real traffic characteristics, while only quantitative results on measured traffic in private DCNs are given and not the actual raw data. In specific, the published results on realistic DCN traffic traces almost exclusively come from Microsoft [68], [71], [72] and Facebook [69], while they refer only to their proprietary cloud applications. The lack of realistic DCN traffic patterns makes hard the effective DCN topology design, since it has to take into account the real traffic conditions while it is heavily dependent on the specific applications running in the DC. The DCN topology design is beyond the scope of this review.

In 2010, Microsoft first published quantitative results on measured traffic in its DCs, while it classified the DCs in three categories: university campus DCs, private enterprise DCs, and commercial cloud DCs, each running different applications [68]. Specifically, in university campus DCs, applications like e-mail, web services, storage back-up and multi-cast video are running, while the dominant application

layer protocol is HTTP. Depending on the specific application, the ratio of DC flows that is served by servers of the same rack (intra-rack traffic) is dynamic, fluctuating from 10% - 40%. In private enterprise DCs, along with the above applications, other specific custom-based applications and development services are running, while the dominant application layer protocols are HTTP, HTTPS, LDAP, and DataBase coming from MapReduce framework. Similar to the university DCs, in private enterprise DCs the ratio of intra-rack traffic varies from 10% - 40%. Finally in commercial cloud DCs, webmail, search engines, and big-data computations like MapReduce-style applications are hosted [68], [71], [72], while the same application layer protocols as in private enterprise DCs are observed. Unlike the previous DC categories, in commercial cloud DCs which serve cloud computing, the vast majority (up to 80%) of traffic is consumed in a single rack, i.e. as intra-rack traffic. Beyond Microsoft results, in 2015 Facebook also published quantitative results about measured traffic in its own DCs, adding one more DC category in the above DCs classification, based on emerging social media applications: the social networks DC category [69].

Although modern DCs host new and more complex types of applications with multiple application layer protocols as compared to the above ones, no more quantitative or qualitative results on measured real traffic in existing DCNs have been published. Therefore, in order for the research community to explore the effectiveness of modern DCN design solutions under more real traffic conditions overpassing real traffic patterns ignorance, synthetic traffic datasets have been used, either generated by simulators [73], [74], [75], [76] or by small-size experimental DC testbeds [77]. However, these synthetic traffic datasets are often unfruitful since they cannot emulate real traffic of high burstiness, heterogeneity and variability that only emerges at existing large-scale DCs.

The efficient DCN design has to take under consideration the following traffic characteristics, which are discussed in the next paragraphs: the traffic locality, the flows size, duration and arrival rate, the size and interarrival time of the respective packets within flows, the traffic burstiness, the traffic priority, the links utilization etc.

1) LOCALITY

Concerning DCN traffic locality, Cisco has reported that almost 75% of the traffic that enters a DCN is served by the DC resources and it is consumed within the DC, while almost 8% is directed to another DC and 17% is sent back to the end-user [78]. In other words, almost three quarters of the traffic that enters a DC follows the east-west communication scenario, as opposed to the rest one quarter that follows the north-south communication scenario exiting DCN after completion. This trend, which remains approximately the same with the traffic increase [79], requires high DCN performance to serve the ever-increasing traffic demands and to provide ensured low end-to-end latency response.

Within a DCN, traffic locality also depends on the specific running applications. As previously discussed based on

Microsoft reports, the intra-rack to inter-rack traffic ratio varies between 10% - 90% and 40% - 60% for university and private enterprise DCs, while it reaches 80% - 20% for commercial cloud DCs [68]. In the latest case, the high intra-rack to inter-rack traffic ratio is because DC operators of cloud storage applications locate into the same rack those servers that usually exchange high amount of traffic between them.

The dependency of traffic locality on the running DC applications has also been verified by Facebook, although it reports that the vast majority of traffic stays within the intra-cluster domain instead of the intra-rack one. In specific, for Hadoop 80.9% of traffic is exchanged inside a cluster, while for web and cache services the respective percentages are 56.3% and 13% [69]. On the other hand, the traffic exchanged in the intra-rack domain is significantly less; 13.3% for Hadoop, 12.1% for web services and 0.2% for cache services. The overall DC traffic locality results are presented in Table 2.

2) FLOW SIZE, DURATION AND ARRIVAL RATE

Regarding flow size, measurements in Microsoft DCs show that the vast majority (more than 80%) of flows have considerably small size (less than 10 KB), while their duration is a few hundreds of ms [68], [71], [72]. There is also a significant flow portion (more than 40%) with size even smaller (less than 1 KB), while the largest flow duration is less than 200 seconds [68]. Other traffic measurements by Microsoft DCs show that 99% of flows have size smaller than 100 MB, while only a small portion of flows have size varying between 100 MB and 1 GB representing 90% of transferred bytes [80]. Very few flows are larger than a few GB [38]. This means that there are few large size flows that contribute with most of traffic bytes often resulting in traffic congestion, while most flows are short enough. In general, a flow is assumed to be short if its size is less than 1 MB [81], [82], [83].

Facebook has also announced statistics about flow size distribution inside its DCs, in regard to the running applications [69]. Hence, for intra-rack traffic, the median and tail flow size is between 100 KB and 100 MB for Hadoop, between 1 KB and 10 KB for cache services, and between 3 KB and 10 MB for web services. The respective flow size for inter-rack traffic is between 1 KB and 1MB for Hadoop, between 30 KB and 3 MB for cache services, and between 5 KB and 500 KB for web services. Concerning the median size flow duration for each separate application in Facebook DCs, it is measured 300 ms for Hadoop, 400 s for cache services, and 900 ms for web services [69]. The overall results about measured flow size and duration in existing DCs are presented in Table 3.

The close relation between flow size distribution and the specific DC application that generated each flow can also be observed in more recent measurements in DCs that serve transport control protocol (TCP)-based applications. For example, real-time interaction cloud applications such as web services, produce flows of varying size between 100 KB and 1 MB for data messages and between 2 KB and 20 KB

TABLE 2. DC traffic locality.

Cisco [78] ¹		Microsoft [68] ¹		Facebook [69] ¹		
Traffic Direction	Traffic	DC Category	Intra-Rack Traffic	DC Application	Intra-Rack Traffic	Intra-Cluster Traffic
Inside DC	75%	University	10% - 40%	Hadoop	13.3%	80.9%
To other DCs	8%	Private Enterprise	10% - 40%	Web Services	12.1%	56.3%
To end-user	17%	Commercial Cloud	80%	Cache Services	0.2%	13%
				Total	12.9%	57.5%

¹ Based on real traffic measurements in existing large-scale DCNs [68], [69], [78].

TABLE 3. DCN flow size & duration.

Microsoft [68], [72] ¹				Facebook [69] ¹				
Flow Percentage	Flow Size	Flow Duration	ToR Traffic Charact.	DC Application	Locality	Flow Size	Median Flow Duration	ToR Traffic Charact.
40%	< 1 KB		ON/OFF	Hadoop	Intra-Rack	100 KB – 100 MB	300 ms	Non ON/OFF
80%	< 10 KB				Inter-Rack	1 KB – 1 MB		
≈ 100%	< 10 MB			Web Services	Intra-Rack	3 KB – 10 MB	900 ms	
Few	100 MB - 1 GB	50 s – 200 s			Inter-Rack	5 KB – 500 KB		
Very Few	> 1 GB	(for large flows)		Cache Services	Intra-Rack	1 KB – 10 KB	400 s	
			Inter-Rack		30 KB – 3 MB			

¹ Based on real traffic measurements in existing large-scale DCNs [68], [69], [72].

for short queries, while they require to be served with high priority [84]. Moreover, big-data computing applications like those of MapReduce framework produce flows of larger size, i.e. between 1 MB and 100 MB [85]. Finally, cloud storage applications provide long-running background large flows of few GB to deliver large amount of data, such as large files [85].

Similar to the flow size distribution, flow interarrival time distribution also depends on the specific application that generated the flow. For example, in Facebook DCs 100 - 500 flows per second have been measured with average interarrival time per server 2 ms – 10 ms [69]. In addition, in Microsoft DCs 100000 flows per second in the intra-cluster DCN have been measured [71]. According to the DC category, measurements at university and private enterprise DCs have shown that the number of flows arriving to a ToR switch varies between 100 and 10000 flows per second. Finally, Microsoft have noted for the aggregated traffic which a ToR switch transmits in an inter-rack DCN that it has ON/OFF characteristics, while both ON and OFF periods duration obeys heavy-tail distributions [68]. On the contrary, Facebook has not reported ON/OFF traffic pattern at the ToR level but only per destination level due to the large number of concurrent destination servers [69], as Table 3 presents.

It is obvious that the measurements of Microsoft and Facebook DC traffic -dated more than a decade ago- cannot precisely describe the present traffic patterns in modern DCs, since new types of cloud applications (like ML/AI, IoT etc.) have emerged meanwhile, definitely providing differentiated

traffic patterns. The lack of new openly accessible real traffic measurements has led some researchers to produce synthetic traffic datasets, either by simulations or by small-scale experimental testbeds, in order to explore DCN optimization solutions. A great challenge for such synthetic datasets is to manage to emulate in the most realistic way the real DCN traffic patterns. To the best of our knowledge, there are few simulation-based DCN traffic generators presented in literature. For most of them, the scope is to feed with generated traffic the comparison study among diverse DCN solutions rather than to be realistic, due to the simplistic enough Poisson flow arrival process assumed [74], [75]. On the contrary, a synthetic traffic generator that follows real DCN traffic patterns has been recently presented in [73], producing traffic of several DC applications under any DCN topology and scalability level. Apart from the already mentioned DC categories, authors of [73] introduce an additional one that is specifically assigned to the emerging ML/AI-based applications and for this reason they upgrade the knowledge about relative DCs traffic patterns. In specific, it is shown that 75% of transferred flows have size smaller than 10 KB inside university DCs, private enterprise DCs and commercial cloud DCs, conforming to previous Microsoft's measurements as Table 3 shows. Regarding social networks DCs and ML/AI applications DCs, simulation results have proved that flows are longer. This is understood since the relative applications exchange larger data volumes. Thus, the size of 75% of transferred flows is calculated to be less than 300 KB and 55 MB in social networks DCs and ML/AI applications

TABLE 4. DCN flow size & interarrival time.

Simulation Testbed DCN [73] ¹			
DC Category	Flow Percentage	Flow Size	Flow Interarrival Time
University	75%	< 10 KB	< 6 ms
Private Enterprise	75%	< 10 KB	< 0.5 ms
Commercial Cloud	75%	< 10 KB	< 0.04 ms
Social Network	75%	< 300 KB	< 1 ms
ML/AI oriented (Tensorflow)	75%	< 55 MB	< 12 ms

¹ Based on DCN traffic datasets generated by simulation [73].

DCs respectively [73]. Finally, concerning the interarrival time of 75% of flows, it is calculated to be less than 6 ms, 0.5 ms, 0.04 ms, 1 ms, and 12 ms for university DCs, private enterprise DCs, commercial cloud DCs, social networks DCs, and ML/AI applications DCs respectively [73]. The flow size and interarrival time in the above simulation results are summarized in Table 4.

3) PACKET SIZE AND INTERARRIVAL TIME

In some research works, it is challenging to explore statistics on a per-packet basis, instead of a flow-based investigation. According to Microsoft DC measurements concerning the respective packets size within flows, it is measured to follow bimodal distribution, while the size of most of the packets is 200 Bytes and 1400 Bytes [68], [71], [72]. These packets correspond to either short control messages of small size (200 Bytes) for admission control that are latency-sensitive requiring immediate response, or long messages of large size (1400 Bytes, i.e. almost equal to the Ethernet maximum transfer unit of 1500 Bytes) as part of large files data exchange that are throughput-sensitive requesting for high bandwidth. The percentage of packets that have size less than 200 Bytes depends on the specific DC category and varies between 30% - 50%, while the rest packets have size around 1400 Bytes [68]. Concerning the packet interarrival rate according to the DC category, it follows the weibull distribution in university DCs and the lognormal distribution in private enterprise DCs [68]. The packet size and interarrival time results are presented in Table 5.

Latest measurements of real traffic in experimental small-scale DC testbed of leaf-spine architecture with 16 servers have updated the packet parameters' assumption for different DC applications [77]. Especially, the packet size and interarrival time has been measured for some of the main cloud application classes: web services, media streaming, Hadoop and cloud storage. As Table 5 shows, for web services more than half of the packets are short enough (less than 200 Bytes) since they correspond to the large amount of control messages that are exchanged to establish web connections. On the contrary, the vast majority of packets

coming from the rest of the examined applications are longer with size almost equal to 1500 Bytes. This is because these applications exchange large amounts of actual data, as compared with the few control packets that are necessary for their connection control. Regarding the packet interarrival time, for media streaming applications it is low enough (less than 12 μ s for 70% of packets), since they exchange real-time data requiring immediate service. On the other hand, cloud storage packets show larger interarrival time since they exchange data of low time-sensitivity and priority. Finally, for the vast majority of Hadoop packets the interarrival time is low enough (less than 8 μ s for 90% of packets). The above experimental measurements are summarized in Table 5.

4) BURSTINESS

Since the prevalent transport protocol in cloud/edge DCs and generally in Internet is TCP due to the dominance of web services, e-mail, file transfer etc., DCN traffic often has highly bursty characteristics. Burstiness of TCP traffic is caused by transport control procedures in software when scheduling a large data window transmission at once like TCP slow start [86]. Moreover, inherited hardware offloading features like Large Send Offload (LSO) and Microsoft Interrupt Moderation may contribute to high burstiness in DCNs, by reducing CPU utilization at high rates and processing packets in batches [87].

Bursty traffic is responsible for link congestion at the top-of-rack (ToR) switching tier of a DCN (see Fig. 4, in Section III), while traffic burstiness fluctuation is becoming smother when moving to the higher DCN switching tiers of aggregation and core [72]. Finally, highly bursty traffic causes high buffers occupancy and consequent high dropping rate and data loss probability at the DC resources. As the level of traffic burstiness increases, both the total packet latency and task completion time increase too [14], [88].

Based on the aforementioned discussion in paragraphs I) to 4), it is concluded that existing DCNs traffic matrixes have high diversity, fluctuation and burstiness while they are almost unpredictable. This fact makes hard the traffic engineering and transmission scheduling within existing DCNs, restricting the achievable level of performance improvement. Knowledge of real flows characteristics such as size and arrival rate can help DCN engineers to design effective MAC and switching schemes.

5) LINK UTILIZATION AND CONGESTION

A significant DCN parameter that plays a pivotal role to the effective DCN design is the links utilization. According to measurements in its DCN topology, Microsoft has reported that link utilization in the core DCN switching tier (see Fig. 4, in Section III) is higher than in the lower aggregation and ToR DCN switching tiers, while it decreases when moving from a higher to a lower tier [71], [72]. Despite low link utilization, congestion -expressed by the loss ratio- is noticed at the lower DCN tiers since congestion is mainly affected by traffic burstiness [72].

TABLE 5. DCN packet size & interarrival time.

Microsoft [68], [72] ¹				Test DCN [77] ²				
DC Category	Packet Percentage	Packet Size	Packet Interarrival Time Distribution	DC Application	Packet Percentage	Packet Size	Packet Percentage	Packet Interarrival Time
University	38% - 50% 45%	< 200 B ≈ 1500 B	Weibull	Web Services	53.1% 37%	< 200 B ≈ 1500 B	81% 10%	< 12 μs < 8 μs
Private Enterprise	30% - 38% 45%	< 200 B ≈ 1500 B	Lognormal	Media Streaming	3% ≈ 96%	< 200 B ≈ 1500 B	90% 4%	5 μs - 25 μs < 5 μs
				Hadoop	8% ≈ 90%	< 200 B ≈ 1500 B	90% 5%	< 8 μs 8 μs - 25 μs
				Cloud Storage	10% ≈ 87%	< 200 B ≈ 1500 B	80% 50%	< 1.2 ms < 100 μs

¹ Based on real traffic measurements in existing large-scale DCNs [68], [72].

² Based on real traffic measurements in experimental small-size DCN testbed [77].

Based on the aforementioned DCN traffic characteristics, some DCN design directions can be outlined. In specific, since the lower ToR switching tier is more affected by traffic congestion due to traffic burstiness [72], extra attention should be paid to the ToR level switching design. This direction is investigated in Section III, where switching technologies and optical switches prototypes are discussed. In addition, since most of the DCN traffic consists of short flows and most of the traffic volume is transferred by few long flows, while packets size distribution varies according to the specific cloud application, the bandwidth allocation and packets transmission scheduling has to effectively take under consideration the specific applications requirements such as latency-sensitivity, tolerance to data losses etc. This direction is extensively explored in Section IV where diverse bandwidth allocation/access control schemes are explored for optical DCNs in both intra- and inter-rack domain.

C. MAIN DCN DESIGN OBJECTIVES

It is worth noting that a DCN does not only interconnect servers within the DC domain, but it is also responsible for the orchestration of all network operations for data transmission, switching, and on time data delivery in general. High data-intensiveness and traffic diversity put in the agenda the following demanding objectives for an effective DCN design, as Fig. 3 presents:

1) PERFORMANCE EFFECTIVENESS

Data-consuming cloud/edge applications running in DCNs set tight performance requirements in order to guarantee a deterministic data delivery, in terms of low and bounded end-to-end latency in the order of few μs, low jitter in the order of ns, high DCN throughput (aiming to reach links capacity), and extremely low packet loss and dropping rate, while the required QoS level strongly depends on the specific application, as previously discussed. To cope with such stringent

requirements, DC operators gradually upgrade the DCN links capacity up to 800G nowadays targeting to the 3.2T era in the near future, as discussed in Section II-A. Also, they incrementally substitute the electrical switching infrastructure by optical switching one providing hybrid electrical/optical solutions in existing DCNs, while the all-optical switching technology is preferred for new DCNs deployment with additional cost, targeting to fast optical switching in the order of ns, as it is discussed in Section III. Finally, it is anticipated that they will gradually adopt effective access schemes to serve the highly heterogeneous traffic in the intra- and inter-rack domain, properly exploiting the increasing DCN link bandwidth, as it is described in Section IV. It is worth noting that the rate of DCN traffic increase is higher than the rate of adoption by DC operators of effective technology solutions, such as optical switching and efficient access schemes, resulting to decreasing DCN performance. For example, Google has announced that during previous years 2004-2012, the bisection bandwidth in its DCs increased 125% per year, since there was available bandwidth to be shared among communicating pairs of servers. This trend has dramatically changed the last years due to the tremendous traffic expansion in Google DCs, reaching only 26% [55].

2) SCALABILITY

The DCN expansion in order to host more processing and storage resources to meet the increasing demands of diverse emerging data-intensive applications is referred as scalability. As a required attribute of a modern DCN, scalability does not just concern the addition of new hardware (CPUs/GPUs and memory) without affecting the existing servers communication, but also require suitable switching and routing protocols to be effectively adapted to the expanded DCN domain. In the light of scalability requirement, switching and access protocols are summarized in Section III and IV respectively.

In existing DCNs, the running cloud applications often determine the DC scale. Hence, commercially available IoT

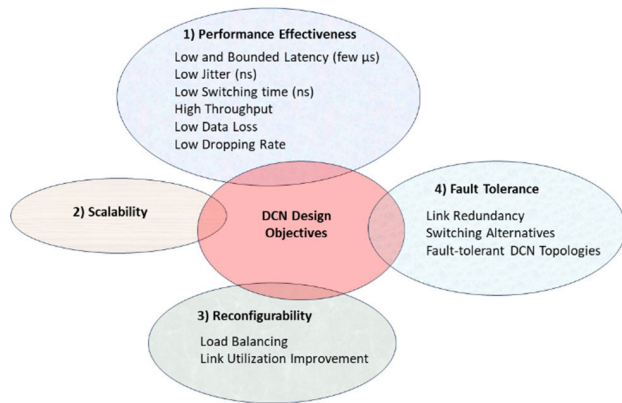


FIGURE 3. Main DCN design objectives.

device control/access services along with relative application platforms, as well as specific custom-based applications are hosted in small-scale DCNs [89] typically at the edge part of the network, benefiting from the 5G technology. Moreover, data-intensive ML/AI applications, cloud computing, cloud storage, and VM migration services are hosted in large-scale cloud DCNs. Since the B5G and 6G era have already risen, dramatically increasing the data volume in DCNs, the demand for new scalable DCN design solutions to enhance the storage and computing resources in a power efficient way becomes pivotal.

3) RECONFIGURABILITY

High traffic dynamicity and burstiness in DCNs impose real-time network resources reconfiguration in order to guarantee the required QoS level for each application [90]. As previously discussed, within a DCN, link utilization varies depending on unpredictable offered traffic conditions. Hence, there may be links between servers, racks or clusters overloaded due to unpredictable data-consuming communication resulting to low unacceptable QoS level, while other links may be underutilized. Thus, efficient DCN resource reconfiguration mechanisms is required to be adopted for load balancing and link utilization improvement, targeting to effective delivery of required QoS level. Time granularity is an important parameter when scheduling this target achievement. In specific, as measurements show in Microsoft DCNs, traffic pattern remains stable in a time scale of tens of ms, while it is dramatically dynamic in a time scale of hundreds of ms [68]. This convinces that DCN resources are required to be fast reconfigurable in the order of few ms.

Network traffic management schemes have been proposed based on linear programming [91], or intelligent algorithms [92], [93]. Their applicability to DCNs is a matter of time scale since duration of reconfiguration processing in the order of seconds or even minutes is unacceptable for real-time DCN traffic. Therefore, heuristic reconfiguration schemes have been adopted, but with limited link utilization improvement [94], [95]. Regarding all-optical DCNs, diverse reconfiguration methods have been introduced too [96], [97],

[98]. Some of them exploit ML techniques for traffic management [99], [100] and deep reinforcement learning (DRL) ones for DCN reconfigurability [77], [101], [102]. In these studies, performance parameters like time duration for model training, DCN traffic dynamicity and burstiness, as well as required cost and latency level achieved should also be taken under consideration. Most of the proposed optical DCN reconfiguration schemes follow the SDN-paradigm for network control [103]. Special investigation has been conducted for optical switches fast reconfiguration to improve DCN performance [9]. Particularly, optical switches based on electro-optical effects achieve ns-scale reconfigurability, while the use of semiconductor optical amplifiers (SOAs) along with optical switches provides lossless and fast switching in the order of ns, like in [104], as it is extensively discussed in Section III-B. Finally, network virtualization mechanisms can be enabled for the efficient DCN operation and management, where DC resources such as network capacity, storage and memory are virtually assumed to slice DCN, targeting to cloud applications stringent requirements satisfaction with limited energy footprint [105].

4) FAULT TOLERANCE

Under often traffic congestion conditions, possible DCN infrastructure failures have to be timely resolved -if not predicted and overcome in advance- in order to support uninterrupted DCN operation. Hence, redundant physical links can be used for traffic redirection in case of link failure. Moreover, fault-tolerant switching schemes can be applied, such as hybrid circuit/packet switching approaches for electrical/optical DCNs improving network resilience [106]. In any case, the avoidance of a single point of failure condition is pursued [107]. Finally, suitable DCN topologies can be adopted to provide DCN robustness [108].

III. SWITCHING IN DCNs

A. ELECTRICAL SWITCHING

Traditional DCNs are equipped with electrical switches in a hierarchical topology of (usually) a three-tier switching escalation (ToR, aggregation and core switching tiers), as Fig. 4 shows. Servers are organized into racks, while each of them hosts some tens of servers. Each server is connected through an electrical copper cable or optical fiber to the ToR electrical switch for further inter-rack communication. ToR switches are connected through optical fibers to higher tier electrical aggregation switches, which are further connected through optical fibers to higher tier electrical core switches. This means that in the higher switching tiers data are transmitted in the optical domain while they are processed and switched in the electrical domain, incurring O/E/O conversions. Such a DCN architecture based on electrical switches suffers by high power consumption and bandwidth bottleneck at higher tiers, while this trend increases as DC scales-up [109].

Regarding traditional DCs power consumption, it is analog to the electrical switch port data rate since electrical switches

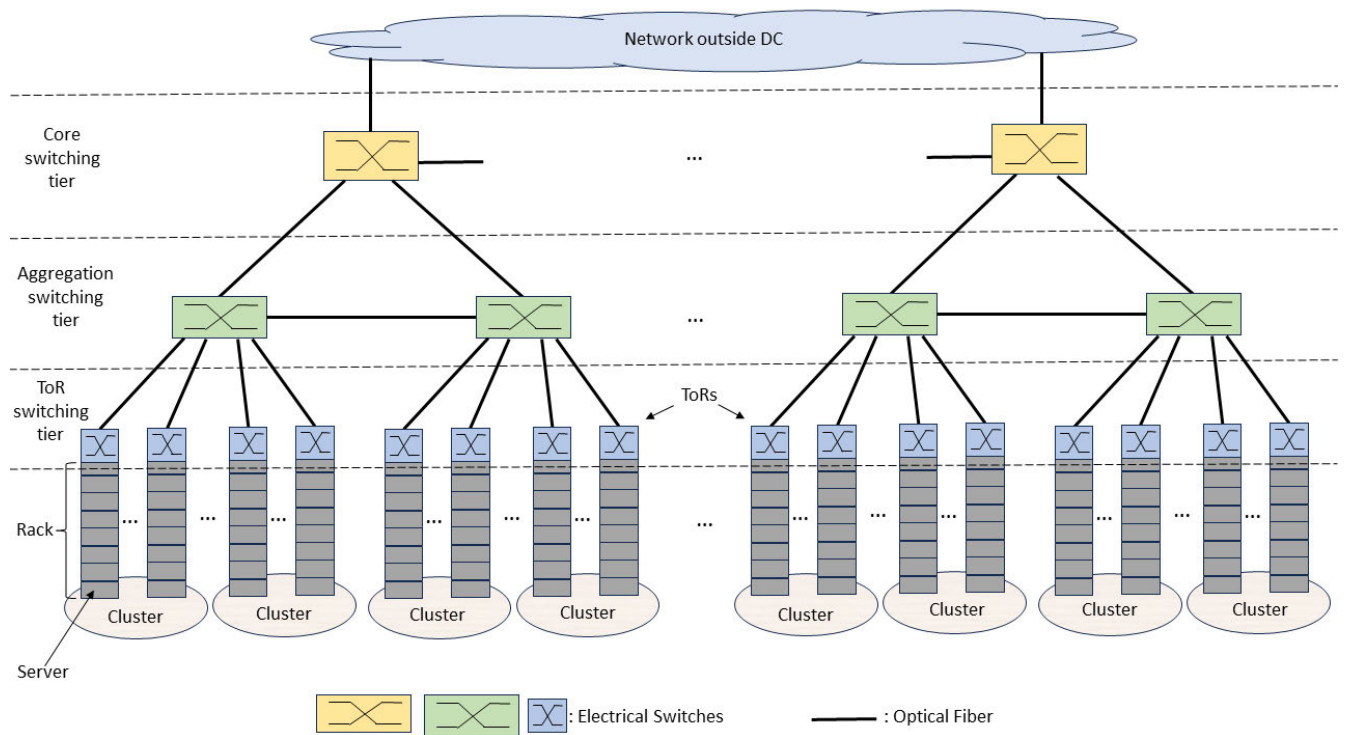


FIGURE 4. Traditional three-tier electrical switched DCN.

consume energy for each switched bit [110]. In specific, hyperscale DCs consume almost 5% of the total power consumption universally [111], [112], while it is expected to dramatically increase during the next years reaching almost 30% by 2030 [112], [113]. Especially for DCNs global electricity demand is expected to reach 3000 TWh by 2030 [114]. Since hyperscale DCs capacity is expected to triple by 2028 [115], it is more than important for DC operators to optimally reduce the energy footprint to serve the accelerated traffic increase. Power consumption in DCs is extensively discussed in Section V.

Moreover, regarding bandwidth bottleneck in current electrically switched DCNs, according to Moore's law it is almost double every two years [55], while it has reached 25.6 Tbps. In the framework of traffic-boosting applications and with DCs scaling-up, this trend is becoming more intensive, especially at the aggregation and core switching tiers due to oversubscription [24]. It is evident that in near future, DCN electrical switch bandwidth will reach unacceptable level of bottleneck due to hardware limitations such as unavailability of switch chip high-speed pins. In order to resolve these limitations, new promising technologies based on silicon photonics, monolithic integration and co-package optics are being explored, despite their immaturity at present [116], [117]. In this line, diverse research projects like COBO [118], OIF [119], and DARPA [120] are investigating the co-packaging optics technology, anticipating energy footprint reduction too.

Therefore, electrical switching is unsuitable to keep up with the traffic-boosting applications in future DCNs in a cost efficient manner. For this reason, optical switching gains ground despite initial deployment significant cost.

B. OPTICAL SWITCHING

Unlike electrical switching, optical switching is transparent to bit rate and data format, providing high flexibility. This is because it does not require any O/E/O conversion at the switch ports since the switching is performed in the optical domain. In addition, it consumes considerably less power per switched bit, as compared to the electrical switching [121]. Moreover, it is worth noting that optical switching provides a more independent relation between the port data rate and the port count (supporting the port count increase without sacrificing the high port data rate), as compared to the electrical switching (which conditions the port data rate and count with the number of interconnected switches). This independence can provide large-scale optically switched DCN topologies avoiding limitations of hierarchical electrical switching such as oversubscription [24].

Based on the aforementioned advantages, optical switching reaches confoundedly higher switch bandwidth as compared to the electrical switching, while it can be improved even more using WDM and SDM techniques [122]. As an immediate result, optical switching can guarantee considerably lower latency and task completion time, while it allows for simpler DCN architectures. For these reasons, optical

switching is expected to be dominant in future DCNs. Table 6 presents an overview of optical switching techniques suitable for optical DCNs.

1) OPTICAL CIRCUIT SWITCHING

To meet the bandwidth-demanding performance requirements of modern DCNs, optical circuit switching (OCS) for DCNs is explored by diverse research projects [43], [44], [123], [124], [125] and implementations [21], while continuous attention is attracted by academia. This is because OCS can provide essential switching bandwidth enhancement with significant power consumption reduction in scalable DCs, providing even two times more cost and power efficient DC solutions [126], as Table 6 presents. Despite the performance limitations discussed in the following, OCS technology is currently preferred by big-data companies' DCs in order to move towards the optical switching era. For example, in 2022 Google replaced the entire spine block by MEMS-based optical circuit switches in order to reduce power consumption and benefit from data transparency [21]. For the OCS switch implementation different optical switching technologies can be used: either MEMS-based [21], [43], [44], [125] or AWG-based [123], [124].

As in circuit switching in classic electrical networks, OCS requires an optical circuit (referred as optical path) establishment prior to the actual data transmission and its release after communication conclusion. A three-way handshake set-up control messaging is required between the source and the destination switch for the optical circuit establishment [127], reserving certain bandwidth for a long time period exclusively for their communication. In this way, no further routing information is needed since all data from the source switch will be routed to the destination switch through the dedicated optical path.

Although set-up procedure provides a pre-defined optical path of guaranteed QoS between the source and the destination switch, it requires long set-up time (in ms order) which may be longer than path actual occupation time for transmission and propagation, especially in highly bursty traffic DCNs. For example for a short burst transmission, the optical path occupation time period (from its transmission by the source switch till its reception by the destination switch) may be much shorter than the required optical path set-up time. Therefore, in such bursty DCN traffic conditions, OCS provides fruitless time consumption for set-up, resulting in low bandwidth utilization. Moreover, the absence of commercially available optical random access memory (RAM) and optical buffer deteriorates the switching performance even more, especially under high burstiness traffic conditions, resulting in higher data dropping probability. In addition, since OCS exclusively assigns a dedicated optical path between two specific switches by avoiding any statistical data multiplexing over it, it considerably restricts the level of maximum bandwidth utilization. Although OCS is widely used in optical networks besides DCs (e.g. reconfigurable optical add/drop multiplexers (ROADMs) are assumed

as optical circuit switches [128]), its applicability to future DCNs should first effectively face the aforementioned challenges in order to handle the bursty and traffic-boosting DCN conditions. Towards this direction, a comparative study of optical switches impact under different OCS integration levels (in core and/or aggregation switching tiers) is given in [129].

2) OPTICAL PACKET SWITCHING

Optical packet switching (OPS) was aspired to be the optical analogue of electrical packet switching (EPS) which has become dominant because of its simplicity in traditional telecom networks and in nowadays electrically switched DCNs. However, OPS technology is not yet a viable solution for DCNs due to the absence of optical RAM for buffering data in the optical domain.

OPS handles DCN traffic on a per-packet basis. According to the OPS functionality, an optical switch -by a packet arrival- extracts the packet header and converts it in the electrical domain (O/E conversion) to process it and define the output switch port according to the destination information carried by the header. Then, the packet payload is routed to the defined switch output port without any O/E/O conversion, providing high throughput and low latency depending on traffic congestion [130], as Table 6 shows. However, an engineering demand arises: the packet payload is required to be properly delayed and stored at a switch buffer as long as its header is being processed for routing purposes. Nevertheless, the immature technology of optical buffers and delay lines cannot support such a functionality yet. Despite technology immaturity, the research community has paid important efforts to adopt OPS in future DCNs by addressing effective contention resolution scenarios [131], [132], [133].

Based on the per-packet transmission scheme, OPS can support statistical data multiplexing over the same wavelength or using WDM technique, allowing for effective bandwidth sharing among different packet flows. Therefore, flexibility can be provided, especially under bursty traffic conditions. In order to meet the OPS stringent time requirements for switching control, fast optical switches in the order of ns switching reconfiguration time have been explored benefiting from the arrayed waveguide grating router (AWGR) [134], [135] and SOA [136], [137] technology, as it is discussed at the end of Section III. In this framework, OPS contention resolution schemes require fiber delay lines (FDLs) or appropriate wavelength conversion with the cost of high power consumption, as presented in Table 6.

3) OPTICAL BURST SWITCHING

Optical burst switching (OBS) has been proposed for DCNs [138], while hybrid OCS/OBS switching schemes have also been introduced [139]. In these cases, AWG/SOA-based optical switching technologies are used for the OBS switch implementation [138], [139].

Similar to the optical path set-up of OCS scenario, in OBS the source switch first transmits a control packet (referred as

TABLE 6. Overview of optical switching techniques for optical DCNs.

Optical Switching Technique	Switching Bandwidth	Switching Granularity	Set-up Time	Total Latency	Header Processing	Traffic Type	Scalability	Power Consumption	Optical Switching Technology	Ref.	Year
OCS	High	Low	High	Low	Low	Long Flows	High	Low	MEMS	[43]	2010
										[44]	2010
										[125]	2013
										[21]	2022
									AWG	[123]	2020
OPS	Depends on Traffic Congestion	High	-	Depends on Traffic Congestion	High	Packets	High	High	AWG	[124]	2020
										[135]	2010
									SOA	[134]	2012
										[136]	2004
OBS	Depends on Traffic Congestion	Medium	Medium	Depends on Traffic Congestion	High	Short Bursts	High	High	AWG/SOA	[137]	2009
										[138]	2016
										[139]	2019

burst header) prior to the actual burst transmission in order to reserve an optical path of predefined bandwidth between the source to the destination switch. By the burst header arrival, all intermediate switches across the path are informed for the burst arrival time in order to properly configure their ports in advance for the burst on-time switching, requiring high header processing delay. Unlike OCS, in OBS the source switch does not require any acknowledgement by the destination switch in order to start actual burst transmission. On the contrary, it waits appropriate offset time in order for the intermediate switches across the path to have configured the switching ports and then it transmits the actual burst. In this way, OBS achieves lower set-up time, while it consequently reaches lower latency and higher bandwidth utilization depending on the traffic congestion conditions. Note that OBS does not necessitates any optical buffering at the intermediate switches, but it requires optical burst contention resolution mechanisms in order to guarantee switch configurability and availability by the burst arrival time to avoid dropping events [140]. Finally, OBS requires the optical path release by the burst flow transmission completion. Since the required offset time period is considerably long, OBS has not met wide acceptance for DCNs especially under high burstiness traffic.

Based on the aforementioned characteristics of OCS, OPS and OBS we can derive useful directions for switching in future DCNs. Thus, although optical switching is preferred to the electrical one in DCNs based on the data rate and format transparency, lower power consumption, simpler DCN architectures and performance enhancement by using WDM/SDM, OPS (as an optical analogue to EPS) is not currently applicable to DCNs due to optical memory lack. Therefore, hybrid EPS and OCS schemes appear as the most promising and practically available switching solution for near future efficient DCNs. In specific, traffic can be categorized according to its size and duration in order to be effectively processed by different switching technology; short traffic flows can be handled by electrical switches, while long flows by optical circuit switches. This approach provides power consumption reduction [141] and high scalability as compared to traditional electrically switched DCN solutions.

The comparison between OCS, OPS and OBS solutions for various aspects of DCN operation is given in Table 6.

4) TECHNOLOGIES FOR OPTICAL SWITCHING

Several technologies have been proposed for optical switching in DCNs. Among them, MEMS, liquid crystal on silicon (LCoS) and beam steering have been commercially available, while optical switches based on MEMS seems to be the most mature technology.

Free space MEMS-based switches can be found either in a two-dimensions (2D) or three-dimensions (3D) implementation. 2D-MEMS switching is based on crossbar topology where micro-mirrors are properly used to steer optical beams to the output switch ports, while 3D-MEMS switching is based on two stages of 2D micro-mirror arrays to steer the optical beams. On the one hand, 2D-MEMS provide short switching (reconfiguration) time in the sub- μ s order, while they provide low cross talk and insertion loss [142], [143], but limited port count up to 64. On the other hand, 3D-MEMS can support large-scale cross-connection [144], up to 576 port counts [145], but they provide long switching time in the order of tens of ms [146]. The support of a few hundreds of switch ports with 3D-MEMS can be realized [147], but it is a matter of cost due to the added complexity. MEMS-based switching technology is the dominant current trend in order to enhance switching capability by saving energy and increasing transparency in existing DCNs. For this reason, Google has recently replaced spine blocks by MEMS-based optical circuit switches in its Jupiter DCN, accepting a few weeks reconfiguration period and few ms switching time [21].

Apart from 3D-MEMS, there are also other optical switching technologies that provide relatively slow switching time varying from ms to μ s. In specific, optical switches based on LCoS technology respond with switching time of tens of ms [147], [148]. Moreover, free space optics switching based on piezoelectric actuators [149] provides switching time in the order of tens of ms too, with commercially available port count of 384. In addition, micro-ring resonator (MRR) technology is used for optical switching using the thermo-optic effect, with switching time of few μ s [150]. Mach-Zehnder

interferometer (MZI) has also been used [151], [152], while MZI-based optical switches is compactly fabricated using silicon photonics technology. Depending on the material, MZI-based optical switches have tens or hundreds of μs switching time for silicon or silica glass correspondingly. Although switching time can be reduced exploiting physical cladding [153] or heater driving [154] dedicated methods, it cannot be lower than a few μs . Further switching time decrease in the order of some tens of ns can be achieved by using silicon photonics and electro-optic effect instead of thermo-optic one [155], [156], [157], while polarization dependencies [158] should be effectively faced.

Moreover, optical switching based on wavelength routing has been proposed since it is simple enough. Indeed, wavelength routing requires wavelength selective (WS) components like AWGs [155], [159] to interconnect switch ports by changing the incoming wavelength to a different outgoing one, often using tunable lasers. In this way, wavelength routing switching performance depends on the tunable laser characteristics; switching time depends on laser tuning time while switching range depends on laser tuning range. Wavelength routing achieves switching time in the order of few ns [160]. Finally, fast optical switching in ns scale has been obtained using SOAs [161], [162], but with limited scalability since it is restricted by amplified spontaneous emission (ASE) noise.

Nowadays, some DC operators gradually adopt free-space optical switches aiming to upgrade switching capability in a long-term schedule. Despite the gradual optical switching integration, the providing switching time by the commercially available free-space optical switches is still in the order of ms, thus short-lived bursty traffic cannot be effectively served. It is obvious that future DCNs have to provide fast optical switching reconfigurability in the order of ns to support data-intensive traffic of high burstiness, providing high scalability. The number of switch ports (as a scalability measure) versus the required switching time is presented in Fig. 5 for the aforementioned optical switches technologies.

IV. BANDWIDTH ALLOCATION IN OPTICAL DCNs

In order to effectively serve the bursty and highly variable traffic in DCs, several bandwidth allocation schemes suitable for optical DCNs have been introduced by academia, for both the intra- and inter-rack DCN domain. It is worth noted that a bandwidth allocation scheme for intra-rack DCNs determines the transmission rules (i.e. intra-rack MAC protocol) followed by the servers of the same rack for their communication, while that one for inter-rack DCNs defines the network resources (i.e. wavelengths, switches ports, light-paths etc.) allocation for the communication among different ToRs. In both cases, bandwidth allocation schemes aim to effectively utilize the available bandwidth determining effective access and switching rules for the communicating servers/ToRs, expecting to reach optimal performance, i.e. low end-to-end latency in the order of few μs , high throughput, and low data dropping rate due to buffers' overflow.

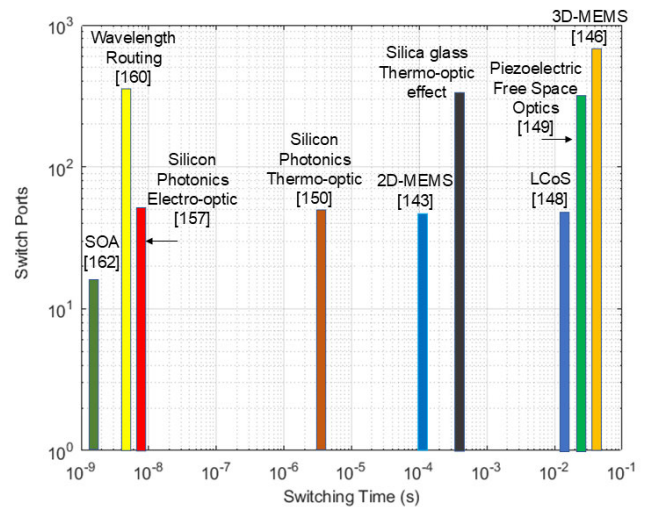


FIGURE 5. Number of switch ports vs. switching time for diverse optical switch technologies.

Finally, bandwidth allocation control is provided either distributedly by each separate communicating server/ToR or centralized by a network central controller implementing the SDN paradigm. Table 7 summarizes the optical DCN MAC protocols for both intra- and inter-rack communication domains.

A. BANDWIDTH ALLOCATION FOR INTRA-RACK COMMUNICATION

Although in most of the existing intra-rack DCNs the communication between servers is implemented over electrical copper cables and the intra-rack switching is performed by electrical switches, there is an intense research investigation about MAC protocols dedicated for optical intra-rack servers' communication, as promising solutions for near future efficient optical intra-rack DCNs [131], [132], [163], [164], [165], [166], [167], [168], [169], [170]. Table 7 (upper part) presents a survey of relative optical intra-rack DCN MAC protocols. As it is shown, in the optical intra-rack DCN domain servers' interconnection is performed by a passive optical interconnect (POI), i.e. either by a AWGR [163] or by a passive optical coupler [164], [165], [166], [167], [168], [169] or by a pair of passive optical combiner and splitter [132], [170], to reduce power consumption. Since the AWGR-based solution is heavily dependent on the specific used wavelengths [163], it essentially restricts network flexibility and effectiveness. For this reason, POIs based on a passive optical coupler or on a pair of optical combiner and splitter are preferred for the intra-rack servers' interconnection to resolve the above drawback [132], [164], [165], [166], [167], [168], since they are wavelength independent.

According to Table 7, MAC protocols for optical intra-rack (and inter-rack) DCNs are categorized according to: 1) the access control scheme (centralized or distributed), 2) the transmission strategy (synchronous or asynchronous), 3) the MAC protocol mechanism for contention resolution

(pre-transmission coordination (PTC), carrier sensing (CS), based on control messages (CM) or based on announcement and confirmation messages (A&CM)), 4) the success of total collision avoidance (yes/no), 5) the offered traffic assumption (packet-based/flow-based), and 6) the offered traffic size and interarrival time distributions.

In specific, regarding the access control in the optical intra-rack DCN domain, it can be performed either distributively by each separate server [131], [132], [164], [168], [169] or centralized by a central controller usually hosted at the ToR infrastructure or at a dedicated central server [163], [165], [166], [167]. In the distributed access control case, each server is responsible to run the MAC algorithm in order to define its own transmission rights according to the mutually followed by all servers access rules. On the contrary in the centralized access control case, the central controller acts as an SDN controller and it is responsible: a) to collect from all servers their control information about their intention for data transmission, b) to centrally run the MAC algorithm based on the received control information in order to define the data transmission rights of each separate server, and c) to announce to all servers the allocated data transmission rights by sending to them appropriate control information. It is evident that the centralized access control scheme suffers from the risk of a single-point of failure [69] and it lacks network flexibility, while it requires longer time to provide data transmission rights to each server. On the other hand, the distributed access control scheme provides a completely server-based bandwidth allocation algorithm running, eliminating the risk of a single-point of failure and mitigating delay overhead [69]. Therefore, although the SDN paradigm is often proposed for next generation optical DCNs for central/administrative control provision, the distributed access control scheme gains ground in latest intra-rack DCN solutions, as Table 7 shows.

Irrespective of the access control scheme, MAC protocols for optical intra-rack DCNs may follow either synchronous [131], [165], [166], [167] or asynchronous transmission strategy [132], [163], [164], [168], [169]. In the former case, it is assumed that time axis is divided in contiguous time periods, called as cycles or slots, while the rack servers can attempt transmission only at the beginning of a time cycle. On the contrary in the latest case, a server may attempt transmission at any time instant. It is worth mentioning that the synchronous transmission strategy may achieve higher throughput as compared to the asynchronous one, since it allows the adoption of efficient MAC protocols such as PTC to totally avoid collisions [131], [165], [166], [167], but it unavoidably necessitates a time synchronization mechanism to be followed by all servers increasing both the capital expenditure (capex) and the operating expenses (opex). On the other hand, the asynchronous transmission strategy is more flexible and less costly since no synchronization mechanism is required [132], [163], [164], [168], [169].

Regarding the contention resolution mechanism proposed for optical intra-rack DCN MAC protocols, the PTC

mechanism has been adopted by both synchronous [131], [165], [166], [167] and asynchronous [163] WDM access (WDMA) transmission strategies irrespectively of the distributed [131] or centralized [165], [166], [167] access control scheme, in order to totally avoid data loss due to collisions. This is because the PTC mechanism allows servers to announce in advance their intention for data transmission i.e. by transmitting appropriate control information prior to the actual data packet transmission. In this way, all servers (for the distributed access control scheme) or the central controller (for the centralized access control one) collect the transmission requests in advance and are able to appropriately coordinate the data transmissions guaranteeing collisions elimination. On the other hand, the contention resolution mechanism based on the A&CM strategy [164] was proposed almost a decade ago for an asynchronous transmission intra-rack MAC protocol, aiming to reduce the number of data collisions but without managing their total avoidance. On the contrary, the contention resolution mechanism based on the effective control messages (CM) exchange among servers prior to the data transmission in order to provide collision-free communication rises as a promising solution especially for asynchronous transmission strategies, despite the nature of the transmission that lacks synchronization and incommodes transmission coordination [168]. Finally, carrier sensing (CS) mechanisms have also gained ground for distributed access control and asynchronous transmission intra-rack MAC protocols. Although they apply effective transmission scheduling strategies in order to reduce data collisions, they reach limited performance due to the unavoidable collisions effect [132], [169]. Therefore, based on the aforementioned discussion about contention resolution, it is concluded that in optical intra-rack DCNs effective contention resolution and total collisions elimination can be achieved by adopting appropriate pre-transmission control coordination mechanisms such as PTC, but paying the unavoidable delay overhead cost.

In addition, key role to the effective intra-rack MAC protocol design plays the realistic assumption of the DCN traffic characteristics. As discussed in Section II-B and especially in Table 3 and Table 5, according to real traffic measurements acquired by Microsoft's [68], [72] and Facebook's [69] DCs as well as according to synthetic DC traffic patterns [77], DCN traffic (packets or flows) is characterized by high variability in interarrival time (providing different levels of burstiness), size, and duration depending on the specific DC applications served. Therefore, intra-rack MAC protocols that are designed to serve traffic of variable size packets (64/1500 Bytes [131], [132], [165], 200/1500 Bytes [163], 200/1400 Bytes [168]) or flows (mice/elephant [166], [167]) adopt a more realistic assumption about the offered traffic as compared to those MAC protocols that assume conventional traffic of a single packet size (1024 Bytes [164], 1500 Bytes [169]), providing more effective performance results. Hence, the consideration of traffic variability in size, interarrival time and duration is of critical importance for the effective intra-rack MAC protocol design and applicability.

Since the vast majority of MAC protocols for optical intra-rack DCNs have been introduced with limited experimental validation, their performance evaluation is mainly based on simulation results. Therefore, the distributions considered for the offered traffic characteristics, i.e. interarrival time, size, and duration, are critical enough for the MAC protocol performance assessment. In specific, the DCN traffic burstiness can be effectively simulated by Pareto distributed ON/OFF transmission periods duration over the network [163], [168]. On the contrary, exponentially distributed traffic interarrival times [131], [132], [164], [166], [167], [169] give a less realistic traffic assumption since they provide conventional traffic patterns that they do not conform to modern DCs traffic variability and burstiness. Finally, other distributions such as lognormal and Weibull can be used to simulate different traffic classes in order to provide services diversity [131], [132]. It is obvious that different traffic priority QoS categories assumption enhances the validation of DCN simulation model results [131], [132] and are preferred for modern DCNs performance evaluation. The aforementioned overview of MAC protocols for optical intra-rack DCNs is given in Table 7.

B. BANDWIDTH ALLOCATION FOR INTER-RACK COMMUNICATION

Unlike optical intra-rack DCNs which mainly use POIs, optical inter-rack DCNs employ optical switches. Inter-rack bandwidth allocation schemes aim to effectively (re)configure the switches resources in order to route the lightpaths carrying the inter-rack traffic from the source to the destination ToR.

The first DCN prototypes employing hybrid electrical/optical switching technology propose MEMS-based optical switches with long reconfiguration time (in μs or ms order) for the optical switching part [43], [44], [171], [172], [173]. Most of them follow the SDN paradigm requiring significant processing overhead with additional processing delay [43], [44], [172], [173]. Therefore, they are only suitable for traffic with low variability such as that of cloud data storage DCNs [172]. On the contrary, all-optical DCN prototypes pursue quicker switching time and lower energy footprint. Some of them use MEMS-based switches reaching switching time in the order of μs [47], [125], while others use WS switches (WSS) with the cost of limited flexibility and latency performance [174], [175], [176]. Moreover, other all-optical inter-rack DCN prototypes employ AWGR-based ToRs interconnection aiming to reach faster performance, but their performance and scalability are limited due to AWGRs dependance on the used wavelengths and port radix [45], [134], [135], [177], [178]. In this framework, Sirius prototype has been introduced by Microsoft research group employing tunable lasers and AWGR wavelength routing mechanisms [124]. Recent DCN prototypes and experimental testbeds exploit the SOA-based optical switching paradigm to provide fast switching time in ns order [92], [179], [180]. Finally, optical passive coupler-based disaggregated

inter-rack DCN prototypes have been introduced in [124] and [181] achieving sub- μs scale synchronous transmission scheduling and up to 93% throughput.

Regarding the MAC protocols and bandwidth allocation schemes for ToR-to-ToR communication in optical inter-rack DCNs, the synchronous transmission strategy is mainly followed [133], [182], [183], [184], [185], [186], [187], [188], as Table 6 (lower part) presents. This means that extra synchronization cost is required to be spent in order for the ToRs transmissions to be appropriately coordinated in the inter-rack DCN domain, totally avoiding collisions [133], [182], [183], [184], [185], [186], [187], [188]. This is achieved by additionally using suitable contention resolution mechanisms such as PTC [133], [182], [183], [184], [185], [186] or routing and wavelength assignment (RWA) algorithms based on wavelength switching matrixes (WSM) [187], [188], as Table 6 shows. Moreover, in the inter-rack DCN domain the optical interconnection among ToRs can be realized using either WSS [182], [183], [184], [185], [186] or AWG [133], [187], [188] reaching low-latency (in μs order) performance. In any case, the ToRs' access control may be performed either centrally following the SDN paradigm [182], [183], [184], [185], [186] or in a distributed way [133], [187], [188], each with the advantages and disadvantages discussed in Section IV-A. Finally, it is worth noting that the extended server-to-server communication along the whole intra- and inter-rack optical DCN path can be also explored, like in [133] and [188] where distributed control, synchronous transmission and WSM contention resolution strategies for diverse traffic priority classes are assumed (while for each case the benefits and drawbacks have been discussed in Section IV-A), reaching low latency in μs order.

Although the aforementioned experimental prototypes of fast optically switched DCNs are still only research-based solutions and not commercially available yet, there is an accelerated interest by both industry and academia about them targeting to even lower optical switching time in the ns scale and lower end-to-end latency in the few- μs scale to enhance the performance of near future optical DCNs.

V. POWER CONSUMPTION IN DCNS

Due to DCN traffic expansion and servers' interconnection escalation, the energy footprint of DCNs is of significant importance for DCs operators, not only because of the high power cost but also because of DCN resilience.

A. POWER CONSUMPTION FOR COOLING

The high density of servers that significantly adds to room temperature, requires power-demanding cooling systems under normal operation even at small-scale DCs. In addition, DCN infrastructure operation itself requires high power consumption, since powerful servers along with electrical switches (still in a big part of existing DCs), active network equipment and highly power-consuming chips in gradually decreasing scales are used. It is worth noting that servers' equipment such as CPU, storage etc. consumes most of

TABLE 7. Overview of MAC protocols in optical DCNs.

Communication Domain	Communicating Nodes	Ref.	Year	Access Control	Optical Inter-connect	Transmission Strategy	Access Contention Resolution	Collision Avoidance	Traffic Interarrival Time Distribution	Mean Traffic Unit Size	
Intra-rack	Server-to-Server (within a rack)	[163]	2017	Centralized	AWG	Asynchronous	PTC ¹	YES	Pareto	200&1500B (Packets)	
		[164]	2017		Passive Coupler	Synchronous			Lognormal	64&1500B (Packets)	
		[165] [167]	2019						Exponential	Mice & Elephant (Flows)	
		[164]	2014	Distributed	Passive Coupler	Asynchronous	A&CM ²	NO	Exponential	1024B (Packets)	
		[168]	2022				CM ³	YES	Pareto	200&1400B (Packets)	
		[169]	2022				CS ⁴	NO	Exponential	1500B (Packets)	
		[131]	2022		M×1 Opt. Combiner & 1×M Opt. Splitter	Synchronous	PTC ¹	YES	Priorities High: Exponential Med: Lognormal Low: Weibull	64&1500B (Packets)	
		[132]	2020			Asynchronous	CS ⁴	NO			
		Inter-rack	ToR-to-ToR	[182] [183] [184] [185] [186]	2018 2019 2018 2017 2018	Centralized	WSS	Synchronous	PTC ¹	YES	
[187]	2019			Distributed	AWG	Synchronous	WSM ⁵	YES	Exponential	Fixed	
Server-to-Server (between different racks)	[133]			2023	Distributed	AWG	Synchronous	PTC ¹ / WSM ⁵	YES	Priorities High: Exponential Med: Lognormal Low: Weibull	64&1500B (Packets)
	[187]			2023							

¹ Pre-Transmission Coordination (PTC).² Announcement and Confirmation Messages (A&CM).³ Control Messages (CM).⁴ Carrier Sensing (CS).⁵ Wavelength Switching Matrix (WSM).

the power in DCs reaching 45% of the total DC power consumption, while network equipment and cooling system consume almost 5% and 40% respectively [189]. In other words, only half of the DC required power is consumed by servers and network infrastructure, whilst a relatively high power percentage is required for the DC cooling. All these power consumption factors are tightly coupled, as noted in [190]. Moreover, the cost of DCN first deployment (buy of network equipment etc.) is less than the power operational cost for cooling and supplying the electrical DCN devices [191].

DCs energy efficiency is measured by the power usage efficiency (PUE) parameter which is defined as the power required for the entire DC operation (e.g. servers, network equipment, cooling, security, lighting etc.) to the power consumed only by the servers and network equipment. In existing enterprise DCs, PUE is close to 1.7 due to the low power consumption optimization for cooling. On the contrary, in large scale DCs such as those of Google and Amazon, cooling power consumption is highly optimized and PUE is close to

1.15 [192], while Facebook has achieved PUE equal to 1.08 in 2022 [193].

In this framework, SotA research about power consumption minimizing for cooling in DCs focuses on air, liquid, and free cooling techniques [194], [195], [196]. More specifically, liquid cooling is used, as by Google for its components [197], while it has been recently applied to hyperscale DCs too, in order to improve power efficiency [198]. Moreover, in order to replace the traditional power-consuming air cooling method, many hyperscale DCs have been placed in areas of cool climate in North America and North Europe to exploit natural cooling. In addition, water or even underwater cooling systems have also been used to reduce energy footprint. Nevertheless, the reduction of power consumption for cooling is not sufficient by itself for DCs total energy footprint decrease since it accounts only 40% of the total DC power consumption. For further reduction, additional improvements is required in energy efficient transmission technologies and network management in the optical domain, as described in the following.

B. ML/AI IMPACT ON DC POWER CONSUMPTION

Since almost 50% of the energy is consumed by servers and network equipment within a DC [189], the advent of ML/AI era provides forecasts for explosive increase that reaches even to 160% of DCs power demands due to the extremely high workloads [199]. In this line, energy demand of DCs serving AI traffic has been forecasted to more than triple during the decade 2020 to 2030, reaching more than 1000 TWh with annual increase of almost 200 TWh [200]. To understand the reason why this will happen, let's think for example for a single ChatGPT query that requires 2.9 Wh as compared with 0.3 Wh required for a Google search, according to International Energy Agency (IEA) [201] and Electric Power Research Institute (EPRI) [202] reports. Therefore, while in existing DCs a rack typically consumes 10-14 kW, this value is expected to rise to 40-60 kW in racks hosting multiple power-hungry GPUs dedicated to serve AI traffic [203]. This is because GPUs consume up to 15 times more power than the conventional CPUs, requiring much more cooling. For this reason, the DCs total power consumption in USA is considered to reach 35 GW by 2030, almost double as compared to 17 GW in 2022 [204]. Similar predictions have been adopted for the 27 countries of European Union (EU-27), according which DCs consumed 1.8%-2.6% of the total energy consumption in EU-27 in 2022, reaching 45-65 TWh [205]. Furthermore, based on the prediction that by 2025 65% of the EU-27 DCs will be hyperscale ones to serve AI traffic, while enterprise DCs will be limited even more and will account for 35% [205], the DCs total power consumption level is expected to rise even more. In this content, big data companies have seen their energy footprint to significantly increase the last years. For example, Meta (Facebook) DCs energy consumption has quadrupled from 2016 to 2020 reaching 7.2 TWh [206], while in Google DCs the energy consumption increase per year was 22% from 2017 to 2021 reaching 18.6 TWh [207]. Similar trends have been observed for Amazon and Microsoft DCs for the same time period [208]. For this reason, renewable energy procurement has already been incorporated mainly from solar and wind energy to move towards sustainable DCs [201].

C. DCN ARCHITECTURE IMPACT ON POWER CONSUMPTION

Under the light of the energy footprint reduction demand, the impact of different DCN architectures on the total DCN power consumption is of crucial importance. In the following, a comparison of some of the most commonly used DCN architectures, such as hierarchical tree-tier, fat-tree, BCube and DCell is provided regarding their energy footprint since they use different number of switch tiers and links among the switches. Table 8 gives the power consumption of the above mentioned DCN architectures as a close mathematical formula [209], taking into account only the power consumed by the switches and the servers that have additional ports, while the power consumption of servers hardware, cooling, lighting etc. is neglected. In Table 8 the parameters E and E_{sw} stand

for the power consumption of a single server and a switch respectively. Moreover, the power consumption is given as a function of the number n of ports that a server or switch can have and the number k of the DCN architecture levels/tiers. For a fair comparison among the different DCN architectures and for the calculation of the power consumption they provide according to Table 8 formulas, commercially available electrical switches have been adopted, while optical switching technology has not been incorporated.

In order to understand the close mathematical formulas of Table 8, let's study separately each DCN architecture. For each architecture, Table 8 also gives -apart from the power consumption- the total number N_{sw} of switches and N_s of servers. First, the hierarchical tree-tier DCN architecture of k tiers has a single switch at the upper tier (core). All switches (both at the upper and the intermediate tiers) have the same number n of ports. The total number of switches in the architecture is N_{sw} given in Table 8, whilst they consume power equal to $E_{sw}N_{sw}$. The servers are hosted at the tree leaves, each connected to a single switch. Therefore, the total number of servers is $N_s = n^k$ while their total power consumption is En^k . Second, the fat-tree topology, also known as Clos topology, has three tiers. The upper tier (core) has $(n/2)^2$ switches, each with n ports [209]. The medium tier has n pods, each with two sub-tiers of $n/2$ switches i.e. each pod has n switches of n ports each. The higher sub-tier switches are connected to the core switches, whilst the lower sub-tier switches are connected to the servers. This means that each pod has n hosts. The total number of servers in the architecture is $N_s = n^3/4$ while their total power consumption is $E(n^3/4)$ [209]. Third, DCell DCN architecture is recursively constructed. The smallest DCell₀ block has a single switch of n ports and connects n servers. In the architecture, switches are hosted only at the lower tier. This means that servers contribute in routing while each of them has $k + 1$ ports. Finally, BCube DCN architecture is also recursively constructed. BCube _{k} block is constructed by n BCube _{$k-1$} blocks and n^k switches each with n ports. Therefore, the architecture can host a total number $N_s = n^{k+1}$ of servers, while the total number N_{sw} of switches is given in Table 8.

Based on the formulas of Table 8, in small-scale DCs of up to 2500 servers, BCube and DCell architectures consume almost the same power. Nevertheless, for large-scale DCs with more than 5000 servers, DCell architecture achieves higher energy efficiency than BCube one. For example, for the BCube architecture with 4096 servers the power consumption has been calculated 73.7 kW, while for the DCell architecture with many more servers, i.e. 5256 servers, the power consumption is much lower, i.e. 55.2 kW [209]. This is because in BCube architecture, servers are not only hosts but they also act as relay nodes for each other, while they can forward traffic between any pair of servers through multiple short paths providing efficient one-to-one and many-to-many communication. For this reason, they require additional network interfaces providing in this way higher power consumption, especially as the number N_s of servers increases.

TABLE 8. Power consumption of DCN architectures [209].

DCN Architecture	Total Number of Switches (N_{sw})	Total Number of Servers (N_s)	Power Consumption
Hierarchical Tree-Tier	$\sum_{i=0}^{k-1} n^i$	n^k	$En^k + E_{sw} \sum_{i=0}^{k-1} n^i$
Fat-Tree	$(\frac{n}{2})^2 + n^2$	$\frac{n^3}{4}$	$\frac{En^3}{4} + E_{sw}[(\frac{n}{2})^2 + n^2]$
DCell	$\frac{1}{n}(n+1)^{2k}$	$\frac{1}{n}(n+1)^{2k}$	$\approx (\frac{E+E_{sw}}{n})(n+1)^{2k}$
BCube	$\sum_{i=1}^{k+1} n^i$	n^{k+1}	$En^{k+1} + E_{sw} \sum_{i=1}^{k+1} n^i$

E : Power consumption of a single server

E_{sw} : Power consumption of a switch

k : number of tiers, n : number of ports per server and switch

Fat-tree DCN architecture power consumption value is calculated between BCube and DCell ones [209], while it depends only on the number n of switch ports, as Table 8 shows. For example, for a 48-port fat-tree DCN architecture that interconnects 27648 servers, the power consumption has been calculated 102 kW [209]. The high power consumption is due to the high power consumed by the ToR, aggregate and core tier switches and the high number of links that are also required. In order to reduce power consumption of fat-tree architecture, the dynamic turning off of switches that are not needed has been proposed providing elastic tree architectures. In this case, even 60% of energy saving is feasible [210]. It is noticeable that regardless the number of servers in the DCN, the power consumption of fat-tree architecture is significantly improved as compared with the BCube and DCell ones when we count in the power consumption of both switches and servers NICs, providing a more realistic estimation of fat-tree architecture energy effectiveness [211].

Finally, hierarchical tree-tier DCN architectures are proven to consume less power regardless the number of servers, as compared with BCube, DCell and fat-tree ones, being the most energy efficient DCN architecture [212]. This is because tree-tier architecture requires less switches at the higher tiers as well as less links. Despite the low energy footprint, hierarchical tree-tier DCN architecture suffers from limited transmission capacity due to traffic bottleneck.

The above results are validated in Table 9 where the above mentioned DCN architectures of electrical switching paradigms are compared regarding their energy efficiency level. Indeed, BCube [213] and DCell [214] architectures provide low energy efficiency regardless the DCN scale, while fat-tree [215] and 3-tier [216] ones provide higher energy efficiency. Finally as Table 9 shows, the Jellyfish [217] and Camcube [218] architectures provide high and medium energy efficiency respectively.

Since an electrical switch consumes energy for each switched bit, its power consumption is an increasing function of its ports data rate [110]. Measurements show that the fat-tree DCN architecture is more dependent on the data rate because of the less links and the low oversubscription,

TABLE 9. DCN architectures comparison – energy efficiency level.

DCN Architecture	Ref.	DCN Topology	Energy Efficiency
BCube	[213]	BCube	Low
DCell	[214]	DCell	Low
Fat-Tree	[215]	Closs	Medium
3-Tier	[216]	Hierarchical	High
Jellyfish	[217]	Flat	High
CamCube	[218]	Torus	Medium

as compared with the other DCN architectures. Specifically, it has been measured that the fat-tree architecture consumes more power than the BCube and DCell ones for all server transmission rates, while DCell architecture is the most energy efficient for all server transmission rate [219]. It is evident that hybrid electrical/optical switching DCNs require even lower power consumption for any server transmission rate [219], since they use energy efficient switching technologies such as MEMS-based optical technology.

Recent research focuses on real-time dynamic modification of optical DCN topology as a function of traffic load and QoS requirements by exploiting the SDN paradigm to minimize the appropriate number of active devices and network paths in order to optimize energy efficiency [220].

D. DIRECTIONS FOR DCN POWER CONSUMPTION REDUCTION

Since optical network technology arises as the dominant one for next generation DCNs, an intensive interest has been invested by both academia and industry to improve their energy efficiency, while guaranteeing high bandwidth, low latency, and high reliability.

1) DEEPENING OF OPTICAL/ELECTRICAL SWITCHING INTEGRATION

Crucial role for the DCN power consumption decrease plays the deepening of optical/electrical switching integration, i.e. the replacement of electrical switches that require power consuming O/E/O conversions by optical switches, with an unavoidable capex increase. The initial predictions were about even 75% of energy saving by transitioning towards the all-optical DCNs that keep signal in the optical domain throughout the network [221]. Energy efficient DCN design approaches adopt the gradual reduction of the number of active components. In this regard, in 2022 Google replaced in Jupiter DC the entire spine block by MEMS-based optical circuit switches, achieving 41% power consumption reduction along with transparency increase [21]. In this direction, hybrid optical switching schemes can be followed in optical DCNs, using on the one hand OCS technology for long data transfers delivering high energy efficiency due to the low circuit reconfiguration needs and on the other hand OPS technology for short data transfers with higher switching and energy demands [222].

2) SILICON PHOTONICS SWITCHING TECHNOLOGY

Given the requirement for energy efficiency in conjunction with low cost and high bandwidth, the technology preferred for large scale optical switch fabrics is that of silicon photonics fabricated in high-volume complementary metal-oxide semiconductor (CMOS) compatible foundries [223], [224]. In silicon-integrated optical switches the MRR- and MZI-based technologies are dominant. They can both use either electro-optic (E-O) or thermo-optic (T-O) phase shift mechanism [225]. The E-O phase shift mechanism reaches higher energy efficiency and switching speed, whilst the T-O one can provide smaller size optical switch [226]. On the one hand, the MRR structure reaches low power consumption, wavelength selectivity and small size. Since it is a mature technology, it has been recently proposed for diverse large scale silicon-integrated optical switches [227], [228]. In opposition to the MRR structure, the MZI one provides higher optical bandwidth whilst it is less sensitive to temperature variations. For this reason it has been introduced in latest studies for silicon-integrated optical switches [229], [230], [231], [232]. In recent research, the technology for silicon-integrated optical switches further incorporating phase change materials (PCM) gains ground, in order to achieve higher energy efficiency in DCs, smaller size and lower insertion loss [233], [234], [235]. The most commonly used PCMs in silicon-integrated optical switches are VO_2 and $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST), while their phase can be changed by heating, electrical and optical triggers [236], [237], [238]. In a silicon-integrated optical switch with PCMs, the structure used is either MRR or waveguide. In Table 10, diverse silicon-integrated optical switches (with MRR [239], or hybrid MRR/MZI [240], [241], or MZI [229], [231], [242], [243]) and silicon-integrated optical switch with PCMs prototypes [244], [245], [246] are presented according to their structural module and phase shift mechanism, while their measured power/energy consumption value is given.

3) ENERGY EFFICIENT OPTICAL TRANSCEIVERS

In order to deliver power efficient transmission in intra- and inter-DC optical links, silicon photonics transceivers technology has been proposed [247], [248], [249]. Also, VCSEL technology is exploited in intra-DC communications since it delivers high power effectiveness. Finally, quantum dot coherent comb lasers [250] and coherent optical transceivers technology with advanced modulation schemes -usually used for long distance communications- can be also used for inter-DC links [251], [252], [253], providing low power consumption.

4) ENERGY AWARE OPTICAL DCN ROUTING, CONTROL & MANAGEMENT

In addition to the above mentioned optical switching and transceivers technologies that deliver reduction of optical DCN power consumption, further energy aware DCN design aspects can be adopted. Specifically, power efficient routing and lightpath management techniques in optical inter-DCNs

TABLE 10. Silicon-Integrated optical switches comparison –power consumption.

Ref.	Year	Structural Module	Phase Shift Mechanism	Power/Energy Consumption
[239]	2012	MRR	E-O	37 mW
[240]	2015	MRR/MZI	T-O/E-O	T-O: 22.37 mW, E-O: 1.38 mW
[242]	2016	MZI	T-O	104.8 mW
[243]	2017	MZI	T-O	117 mW
[244]	2018	Waveguide	PCM	T-E ¹ : 45 mW, T-M ² : 70 mW
[241]	2018	MRR/MZI	T-O/E-O	0.34 mW
[245]	2020	Waveguide	PCM	9.5 nJ and 3.2 nJ
[229]	2020	MZI	T-O	0.48 mW
[246]	2021	Waveguide	PCM	3.528 pJ
[231]	2021	MZI	T-O	22.6 mW

¹T-E: Transverse Electric.

²T-M: Transverse Magnetic.

have been introduced exploiting the SDN capabilities to control power consumption by appropriately avoiding the unnecessary tearing down and reactivation of lightpaths [254]. In other words, energy aware DCN traffic management and traffic aggregation techniques based on SDN approaches can be used to turn off underutilized links, switches and devices during off-peak time periods to improve energy saving. Moreover, network virtualization along with SDN mechanisms have been proposed to virtualize and centrally manage optical DCN functions and dynamically allocate network resources. In this way, inactive DCN parts can be powered off providing energy efficiency, while power saving real-time resource provisioning on demand can be supported reducing power consumption during optical transceivers and switches idle state [255], [256]. In this framework, ML/AI techniques of analyzing DCN traffic patterns are further exploited to predict in advance the upcoming DCN traffic variation in order to deactivate optical links and devices in case of low utilization forecasts, optimizing energy efficiency. For example, DRL techniques are used to evaluate energy consumption in DCNs [257] and to improve power saving in SDN-based DCNs [258], while reinforcement learning techniques are proposed to enhance energy efficient DCN resource allocation [259], by minimizing energy demands based on DCN traffic learning patterns.

VI. FUTURE DIRECTIONS IN OPTICAL DCNs RESEARCH

Despite recent research results and prototypes about many aspects of optical DCNs deployment, there are still open research topics that require further investigation in order to reach high DCN effectiveness to serve the bursty and highly variable DC traffic. Although there are recent optical components technology evolutions that can provide improved optical switching reconfigurability in the ns order, the total performance enhancement of optical DCNs cannot be guaranteed if other networking requirements such as flexibility, scalability and differentiated service of variable and

priority-based DC traffic are not additionally supported. The most challenging open research topics for next-generation optical DCNs as well as their limitations are summarized in the following paragraphs.

A. NS SWITCHING CONTROL

It is worth mentioning that the discussion in Section III-B about recent fast optical switching achievements in the ns scale only refers to the actual fast optical path switching between two specific ports of the optical switch. In other words, it is assumed that the decision about the switching between these specific switch ports has been taken in a previous time instant and it is not counted in the switching time calculation. It is obvious that this decision requires the suitable processing of control information about the destination of the switched data. In order for the total optical switch response time to be restricted in ns scale for fast switching, the switch control for destination information processing must be also performed within a few ns [260]. Up to now, the technology for switching control and required information processing is not mature enough to provide them within a few ns in a scalable and efficient way. This restriction gets worse the realization of optically switched DCNs with packet granularity since they additionally require synchronization in the ns scale. Thus, fast and efficient switching control in ns scale remains an open research request.

B. FAST CLOCK DATA RECOVERY

In order for packet-based bursty traffic optical DCNs to achieve efficient performance in terms of latency, fast CDR within few ns is required to be provided by the optical receivers. If fast CDR is not supported, the total DCN performance deteriorates since throughput reduces and latency increases, especially under short-size bursty traffic conditions as in modern DCNs [261]. Nowadays, CDR in ns scale is achieved by burst-mode receivers [262] only in low data rates. Since near future optical DCNs will enable high data rate operation at 800 Gbps or even higher per wavelength, as mentioned in Section II-A, fast CDR in ns scale for such high data rates is required, being a hot challenge for optical networks engineers.

C. OPTICAL BUFFERING

As mentioned in Section III-B, OPS is not currently a realistic analogue of EPS in optical DCNs since no optical buffering is available due to technology restrictions. Therefore, optically transmitted traffic within an optical DCN cannot be effectively stored in the optical domain upon its arrival at an optical switch, as long as the appropriate control information processing is taking place for switching control. This fact results to high data loss rate and dropping probability especially under highly bursty traffic, as mentioned in Section III-B. Although some alternatives such as FDLs and wavelength conversion techniques have been introduced to resolve the high dropping rate, they have limited applicability in large-scale DCNs due to the increasing complexity and

high queuing delay (that burdens even more the end-to-end latency), respectively. Thus, optical buffering is a pivotal challenge for efficient optical DCNs.

D. BANDWIDTH & ACCESS CONTROL EFFICIENCY

The lack of optical buffering to store packets in the optical domain without requiring O/E/O conversion in conjunction with the absence of fast switching control and CDR as well as with the stringent synchronization requirement especially for bursty traffic service, have a substantial effect on the total performance of nowadays optical DCNs. This is because they impose strict packet contention resolution mechanisms and bandwidth allocation schemes to achieve optimum traffic congestion control and transmission performance due to technology limitations. As already discussed in Section IV, efficient access control and resource allocation schemes aim to provide guaranteed high DCN bandwidth utilization, extremely low end-to-end latency (in a few μ s) and task completion time, while ensuring for low (if not zero) packet loss and data dropping probability, taking into account the traffic burstiness, variability and QoS priority classification in modern DCNs. Therefore, it remains an open research topic.

E. SELF-DRIVEN DCN RECONFIGURABILITY

As mentioned in Section II-C, high reconfigurability and adaptability to the variable traffic conditions is required for next generation optical DCNs, in order for them to efficiently serve the intensive bursty traffic. In specific, this flexibility is required for both the optical switches reconfiguration and the bandwidth allocation schemes on-time adaptability according to the varying traffic demands. ML/AI can significantly support such challenging achievements. In other words, both optical switch and network resources reconfiguration can benefit from ML/AI techniques to provide effective switching and access control and management in the optical domain. In this way, self-driven optical DCN reconfiguration according to the varying DC traffic needs is achieved, avoiding the currently required human intervention.

In this direction, multiple ML techniques have been already developed to enhance DCN decision-making ability and intelligence. They aim to forecast various DCN performance parameters values as a-priori knowledge in order to improve DCN automated reconfiguration ability in case of DCN scaling up and ever-changing traffic conditions which provide high complexity and difficulty in network optimization. This is achieved by adjusting appropriate learning strategies to grasp DCN traffic characteristics in advance and to appropriately reconfigure DCN to provide optimum performance (maximum bandwidth utilization, and minimum latency and data loss), as well as power efficiency. More specifically, DCN design and operation can benefit from such ML/AI-based algorithms about flow prediction [263], [264], flow classification [265], load balancing [266], [267], [268], resource management [269], [270], [271], routing optimization [254], [266], [272], [273], network topology [274], congestion control [258], [262] etc. to optimize DCN

performance. Despite the considerable efforts, there are still many challenges for optical DCNs intelligence improvement, especially for online real-time dynamic reconfiguration ability under heterogeneous traffic and applications.

For the learning model training, suitable previous data knowledge and relative datasets should be acquired. In this framework, the adoption of a knowledge and intelligence higher management DCN plane [276], could be beneficial to determine the optimum reconfiguration scheme for high performance, low latency, low energy consumption, and high fault tolerance. For example, an optical DCN reconfiguration procedure has lately been proposed in [277] exploiting ML techniques based on monitored datasets. Moreover, self-supervised learning techniques have been explored for reconfigurable optical DCNs [278], while transfer learning as a part of self-supervised learning technique has also been investigated [279]. Concluding, although ML/AI is mainly used for DC traffic pattern prediction [280] for optimization purposes, strong emphasis could be placed to use ML/AI to additionally adopt self-driven optically reconfigurable DCNs according to the constantly increasing and highly varying traffic demands.

F. SPACE DIVISION MULTIPLEXING IN DCNs

The explosive increase of traffic in DCNs due to ML/AI dominance has already intensively posed under investigation the demand of expanding DCN capacity. Although WDM technique is currently used in optical DCNs as a well-known technology, the research effort mainly on SDM incorporation in DCNs is expected to continue at pace in the near future to provide more mature technologies to enhance even more capacity. SDM technique is currently used in DCNs exploiting the commercially available fiber ribbons or bundles [281] which can be obtained by packing together thousands of single mode fibers (SMFs) [282]. In this way, high capacity, special efficiency and compatibility with WDM technology can be achieved [283], while cabling complexity is significantly reduced. Despite capacity boost, SDM technique based on fiber ribbons or bundles provides bulky cables which are hard to manage in a dense DCN environment, in order to serve the explosive DCN traffic. To overcome this issue, multi core fibers (MCFs) [284], [285] and few-mode fibers (FMF) [283] can be used in DCNs implementing the SDM scenario. Although MCF capacity is linearly proportional to the number of cores in the fiber, MCF effectiveness is restricted by the inter-core crosstalk (IC-XT), i.e. the undesired high coupling between optical signals that propagate in adjacent cores and can occur if the principal and the undesired IC-XT signals match in phase [286]. In this case, optical signal-to-noise (OSNR) decrease is provided resulting in high power demand [287]. For this reason, only uncoupled MCFs [288] and weakly coupled MCFs [289] can be exploited in short-reach applications such as DCNs, since strongly coupled MCFs [290] provide high IC-XT even in a few meters fiber length [291]. In a MCF, the core number increase (in order

to reach higher capacity) results to MCF cladding diameter increase too. This fact is critical enough for DCNs design where high special efficiency is required. Although some efficient MCF designs suitable for DCNs have been proposed (e.g. a MCF with 100 cores within a 320 μm cladding diameter, providing IC-XT < 25 dB for 1 km fiber length [292], and another MCF demonstration of more than 100 spatial channels each carrying hundreds of WDM channels in a single fiber with capacity > 10 Pbps [293]), there is still an intensive interest to achieve even higher core density and lower IC-XT, in smaller cladding diameter and for longer fibers for efficient DCN implementations. Nowadays MCF-based SDM technology is assumed to be the most efficient and feasible one to increase capacity for short reach DCNs, since IC-XT can be considered not severe over short links (< 1 km). Since MCF-based SDM technique is achieved to be compatible to energy efficient integrated technologies such as VCSELs [294], [295], while it can be used with silicon photonic on-board transceivers [296], it is a promising candidate to meet the demands for high capacity, cost and power efficient DCNs despite the short-reach implementation restriction and the consequent small DCN scale. In this direction advocates the evolution of optical switches that can support MCF links with SDM switching suitable for DCNs [297], [298], as well as of SDM MUX/DEMUX modules [283]. On the other hand, MCF amplifiers technology is still under research. Finally, efficient MCF-based SDM DCNs can be envisioned with the exploitation of effective transmission techniques such as scheduling transmission in opposite directions (bi-directional transmission) over adjacent cores to suppress IC-XT [299], [300], along with appropriate routing and spectrum allocation (RSA) algorithms for different DCN topologies on the top of SDM and WDM [301], [302] to increase DCN capacity up to even 300 times as compared to the WDM solution.

G. DISAGGREGATION IN DCNs

In order to meet the demands of serving the explosive traffic, DC operators gradually enhance the DC resources total capacity by increasing the number of networking, memory, storage and computing components. However, the memory and CPU utilization in DCs is low enough. Measurements have revealed that DC clusters utilize only 10%-30% of the total CPU capacity during 80% of the time, whilst the average memory utilization is only 50% during 55% of the time [303]. In other words, despite the high cost and power consumed for the constantly increasing number of DC resources, they cannot be fully utilized. This is because traditional DCs follow the server-centric paradigm according which the available resources per server are limited to the boundaries of the mainboard tray. Nevertheless, the ratio of resources demands for memory to those for CPU may vary by 3-4 orders of magnitude, being higher than the available resources in a server tray [304] and providing resource underutilization. For example, if in a server the whole CPU capacity is allocated to a single task along with a part of the

memory resources, then the remaining memory is wasted [305], providing resource fragmentation [306] which accounts for even 85% of the total DCN cost [307]. It is evident that DC resources wastage is getting worse with DC scaling. In order to enhance DC performance, the resource-centric paradigm has attracted research interest, according which server resources (memory, xPU, storage and accelerators) are disaggregated into standalone components [308], [309], [310], [311]. According to the disaggregated DC (DDC) approach, all servers resources provide a common pool of accessible resources across the whole DCN, from which a resource can independently from the others be allocated to a task. In this way, DDCs can offer high flexibility for resource allocation, while resource utilization can be improved even up to 34% and power consumption up to 40% [312]. In the DDC architecture, a great challenge arises: high bandwidth and low latency interconnect network is required to ensure the memory-to-CPU ultra-high speed communication. This is because DDC traffic is exchanged among more nodes (the disaggregated resources) than in traditional direct-attached components DCs providing high traffic congestion, while higher latency overhead is provided. In nowadays computer architecture, the interconnection among multiple resource types poses different latency requirements. For example, the communication between CPU and storage can be implemented in ms scale, while that of CPU and memory must be realized much faster in ns scale [313]. As a promising solution to meet these challenges arises optical networking technology. Although optical transmission technology can support high (but not unlimited) bandwidth in Tbps scale, optical switching technology is not mature yet to provide such low switching latency. Despite recent research prototypes that have met ns scale switching demands [310], [314], [315], [316], the challenge to fully utilize the advantages of fully DDCs in a power efficient way with low latency and high bandwidth still calls for further advances.

objectives and the future research directions of optical DCNs, as they are discussed in the previous sections.

VII. CONCLUSION

ML/AI advent has enabled several new cloud/edge applications that impetuously gain ground in every-day life, such as ChatGPT, requiring enormous datasets to be exchanged and processed in a flash inside DCs globally. Moreover, multiple emerging data-intensive delay-sensitive user applications, such as social networks, online gaming etc., are co-hosted in DCs providing a tremendous traffic explosion within DCNs. These extremely intensive traffic-boosting conditions in existing DCNs drive new tendencies in DCN operation and design in order to support the end-users preferences without additional apparent complexity. Optical technology in transmission and switching is projected to effectively serve the rapidly increasing traffic in DCNs.

This paper gives a comprehensive review on the main characteristics of existing DCNs while it emphasizes the optical technology that enables next generation DCNs to provide rich performance in terms of low end-to-end latency in μ s scale. In specific, this paper presents the recent technology evolution on optical DCN links and it outlines the expected relative evolutions in the near future towards the Tbps data rate era. In addition, it extensively explores the DCN traffic characteristics of high burstiness and variability. Based on both measurements in existing big-data companies' DCs and published simulation DCN traffic predictions, it discusses the close dependance of traffic size, interarrival time and locality on the specific applications running in each DCN, deriving useful directions for efficient optical DCN design. Since, the rapid traffic increase and burstiness in DCNs can be effectively served by optical switching technology as compared to the electrical one that is currently used in most of DCs globally (despite the introduction of few optical switching prototypes), this paper focuses on optical DCN switching. In specific, it provides an extensive overview of the optical switching technologies and techniques that are currently used in existing DCNs and of those which are expected to enable advances in new emerging data-intensive cloud/edge applications in next generation optical DCNs. Furthermore, bandwidth allocation schemes and access control mechanisms have been extensively discussed for both intra- and inter-rack optical DCN domain, aiming to reach efficient contention resolution, high bandwidth utilization and low end-to-end delay. Moreover, power demands of modern DCNs is further discussed, while diverse DCN architectures and optical switching technologies are compared regarding their energy footprint. Finally, this paper outlines the technical challenges and open research topics towards efficient optical DCN performance in the near future. Since multiple data-boosting time-sensitive user applications constantly emerge, optical technology is expected to play crucial role in cloud/edge DCs with adaptability, reconfigurability and intelligence enabled by ML/AI, providing fast and reliable service to the end users, while keeping cost and power efficient DCN operation.

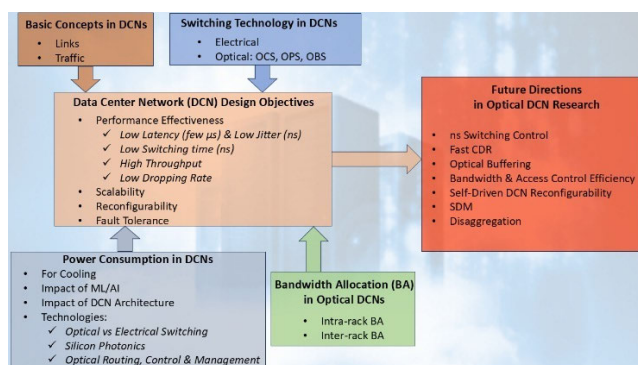


FIGURE 6. Key technologies, basic concepts, bandwidth allocation schemes, design objectives and future research directions of optical DCNs.

Concluding the review study, Fig. 6 presents the interaction diagram that summarizes the key technologies, the basic concepts, the bandwidth allocation schemes, the design

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